AWS JOBS YOU CAN GET WITH AN AWS CERTIFICATION

- #1. Operational Support Engineer
- #2. Cloud Software Engineer
- #3. System Integrator — Cloud
- #4. Cloud Developer
- #5. Cloud DevOps Engineer
- #6. AWS Solutions Architect
- #7. AWS SysOps Administrator
- #8. Senior AWS Cloud Architect
- #9. AWS Consultant

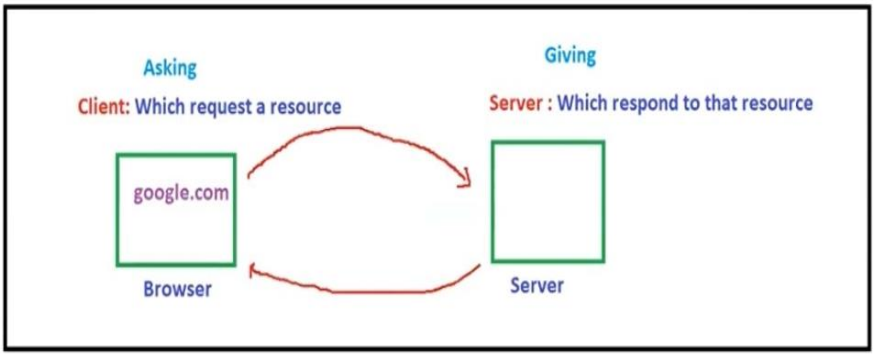
Activate Windows
Go to Settings to activate Windows.

COURSE OUTLINE

Client Server Architecture
Network Basic Concepts
Virtualization Overview
Cloud Computing
Amazon Web Services Overview
High Availability Architecture
AWS Sign UP Procedure
MFA Configuration
AWS CLI
Compute (EC2, Elastic Beanstalk, Light Sail, Lambda,GA)
Storage (S3, EBS, EFS, Storage Gateway, Glacier, Snowball)
Databases(RDS, Redshift, DynamoDB, ElastiCache)
Network and Content Delivery(VPC, Route53, CloudFront, Direct Connect, Global Accelerator)
Management Tools(CloudWatch, CloudTrail)
Security(IAM, Trusted Advisor and Inspector)
Application Services(SWF, Transcoder)
Messaging (SNS, SQS and SES)
Overview on Developer Tools(Code Commit, CodeBuild, CodeDeploy, Code Pipeline)
Route53 Failover Project
Architecting VPC from scratch(Project)
Final AWS project including multiple services

Activate Windows
Go to Settings to activate Windc

Client Server Architecture



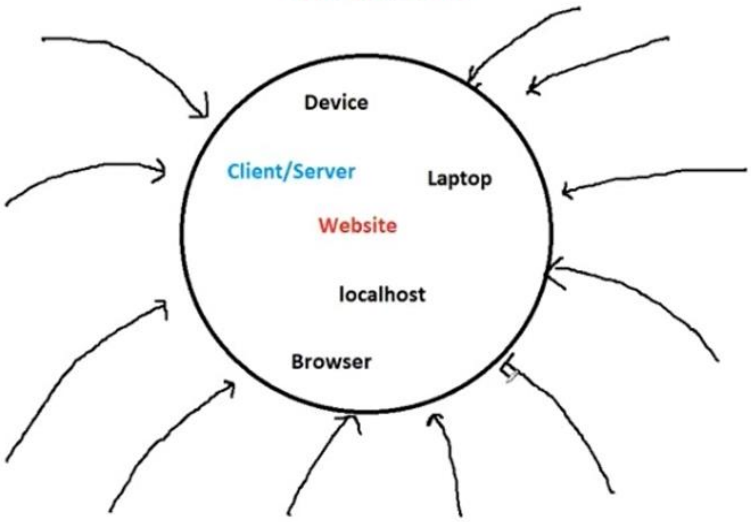
Network

Device can be either a client or a Server

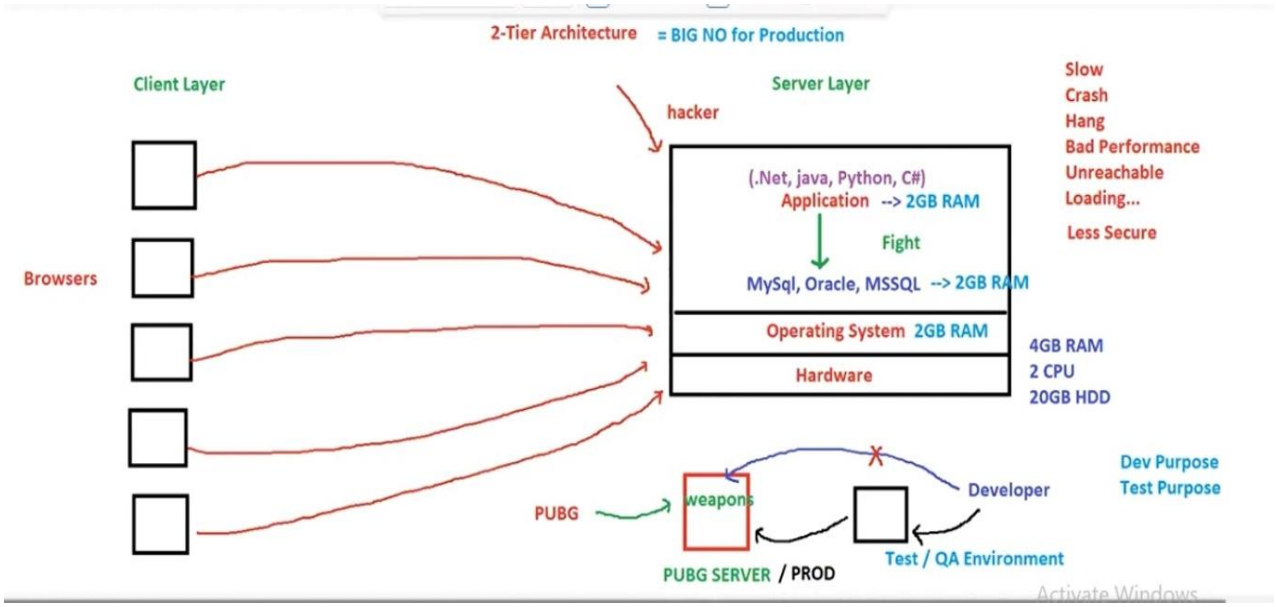
1-Tier
2-Tier
3-Tier

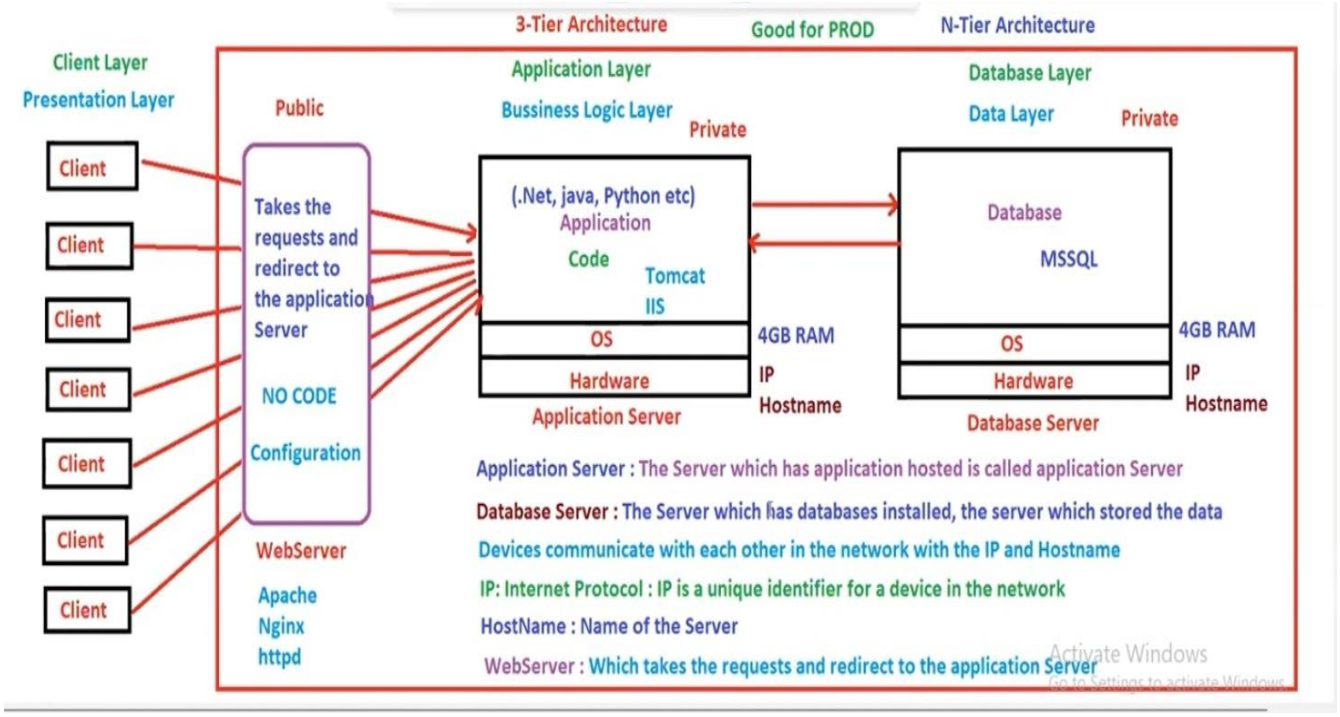
Activate Windows
Go to Settings to activate Windows.

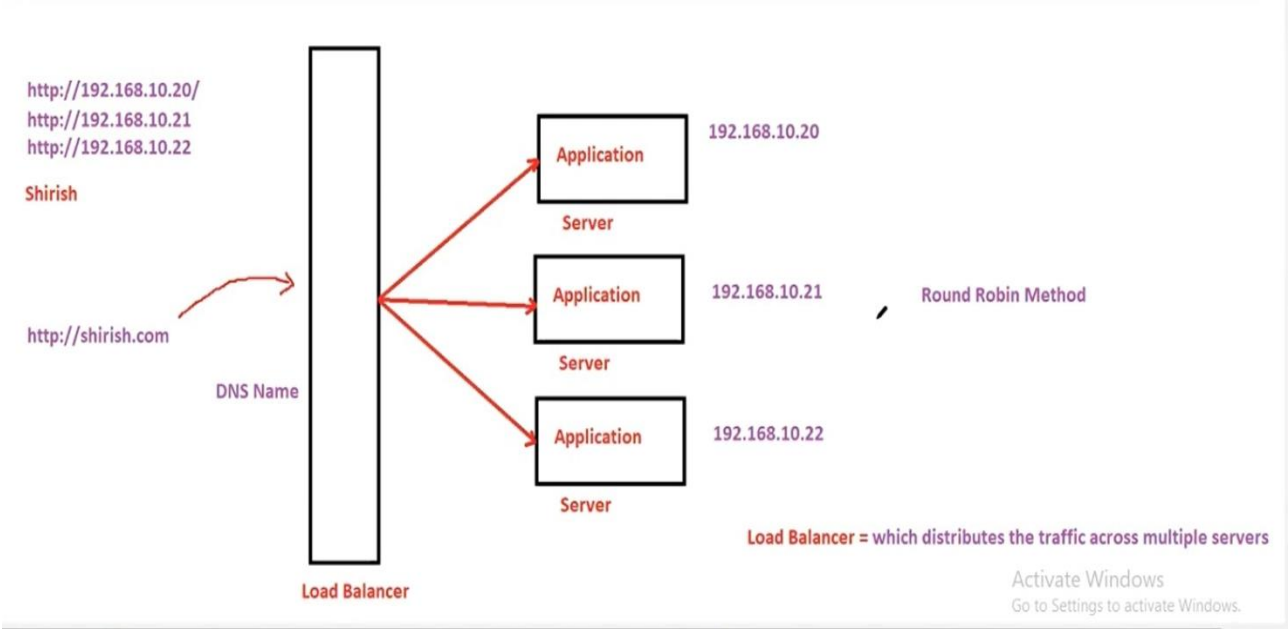
1-Tier Architecture

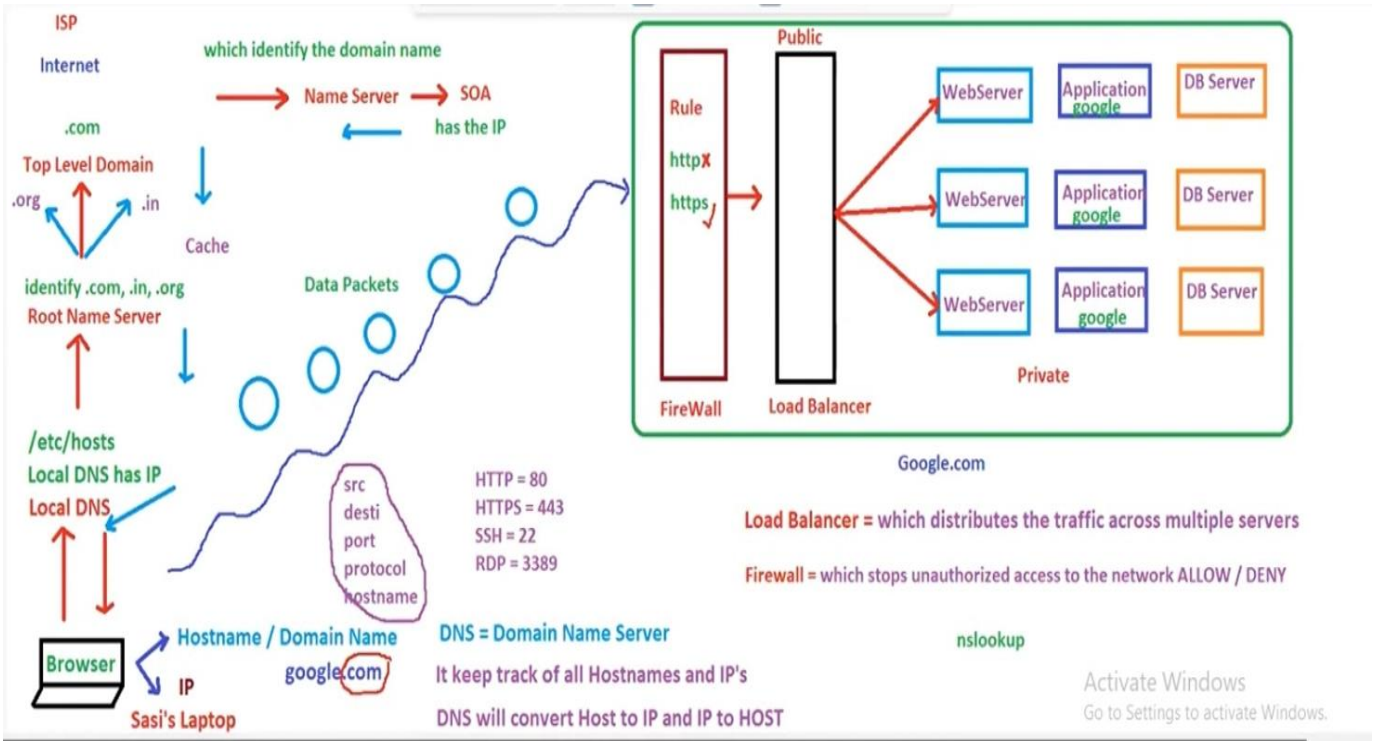


Act


2-Tier Architecture = BIG NO for Production
Client Layer
Server Layer
hacker
Slow
Crash
Hang
Bad Performance
Unreachable
Loading...
Less Secure
Browsers
(.Net, java, Python, C#)
Application --> 2GB RAM
Fight
MySql, Oracle, MSSQL --> 2GB RAM
Operating System 2GB RAM
Hardware
4GB RAM
2 CPU
20GB HDD
Dev Purpose
Test Purpose
PUBG
weapons
Developer
Test / QA Environment
PUBG SERVER / PROD
Activate Windows


3-Tier Architecture
Good for PROD
N-Tier Architecture
Client Layer
Presentation Layer
Application Layer
Bussiness Logic Layer
Private
Database Layer
Data Layer
Private
Public
Client
Client
Client
Client
Client
Client
Client
Takes the requests and redirect to the application Server
NO CODE
Configuration
(.Net, java, Python etc)
Application
Code
Tomcat
IIS
OS
Hardware
4GB RAM
IP
Hostname
Application Server
Database
MSSQL
OS
Hardware
4GB RAM
IP
Hostname
Database Server
WebServer
Apache
Nginx
httpd
Application Server : The Server which has application hosted is called application Server
Database Server : The Server which has databases installed, the server which stored the data
Devices communicate with each other in the network with the IP and Hostname
IP: Internet Protocol : IP is a unique identifier for a device in the network
HostName : Name of the Server
WebServer : Which takes the requests and redirect to the application Server
Activate Windows
Go to Settings to activate Windows.


http://192.168.10.20/
http://192.168.10.21
http://192.168.10.22
Shirish
http://shirish.com
DNS Name
Load Balancer
Application
Server
192.168.10.20
Application
Server
192.168.10.21
Application
Server
192.168.10.22
Round Robin Method
Load Balancer = which distributes the traffic across multiple servers
Activate Windows
Go to Settings to activate Windows.


ISP
Internet
which identify the domain name
Name Server
SOA
has the IP
.com
Top Level Domain
.org
.in
Cache
identify .com, .in, .org
Root Name Server
Data Packets
/etc/hosts
Local DNS has IP
Local DNS
Browser
Hostname / Domain Name
google.com
IP
Sasi's Laptop
src
desti
port
protocol
hostname
HTTP = 80
HTTPS = 443
SSH = 22
RDP = 3389
DNS = Domain Name Server
It keep track of all Hostnames and IP's
DNS will convert Host to IP and IP to HOST
Public
Rule
http
https
FireWall
Load Balancer
WebServer
WebServer
WebServer
Application
google
Application
google
Application
google
DB Server
DB Server
DB Server
Private
Google.com
Load Balancer = which distributes the traffic across multiple servers
Firewall = which stops unauthorized access to the network ALLOW / DENY
nslookup
Activate Windows
Go to Settings to activate Windows.

Browser --> Local DNS --> Root Name Server --> Top Level Domain --> Name Server --> SOA
Browser --> Firewall --> Load Balancer --> WebServer --> Application Server --> DB Server

HTTP = HyperText Transfer Protocol

HTTP default Port Number is 80

You can customize the port number on application level

http://google.com

Protocol://Domainname:PortNumber

http://192.168.10.10:80

http://192.168.10.10:8080 ✓

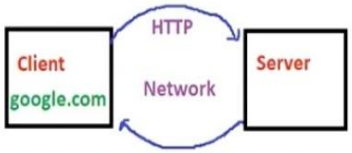
http://192.168.10.10

For customer its always recommend to expose port 80 only

https - 443



Server 192.168.10.10



Status Code
404 = Page Not Found
503 = Service Unavailable
401 = Unauthorized
500 = Internal Server Error
200 = Success Code

HTTP Over TCP/IP

Transmission Control Protocol

HTTP transfers the data to and fro from the browser to Server

HTTPS - 443 (Secure)

Certificates SSL

Encrypted

SSL / TLS / HTTPS



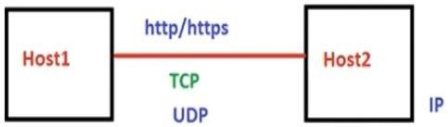
http (Not Secure)

hacker

Plain Text

Data Packet

Host = Client/Server

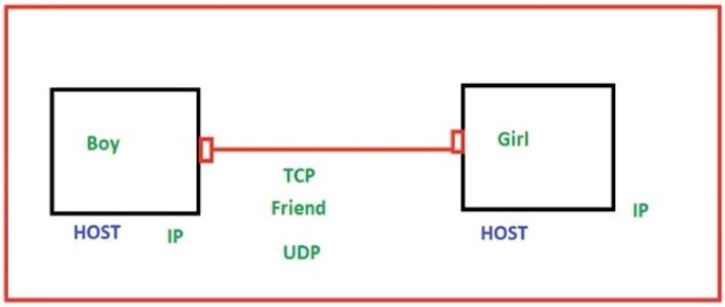


TCP establish the connection between the 2 hosts

TCP is like a messenger / bridge

Activate Windows
Go to Settings to activate Windows.

OSI Layers



7 Layers

Application Layer (http)

Presentation Layer (ssh, rdp)

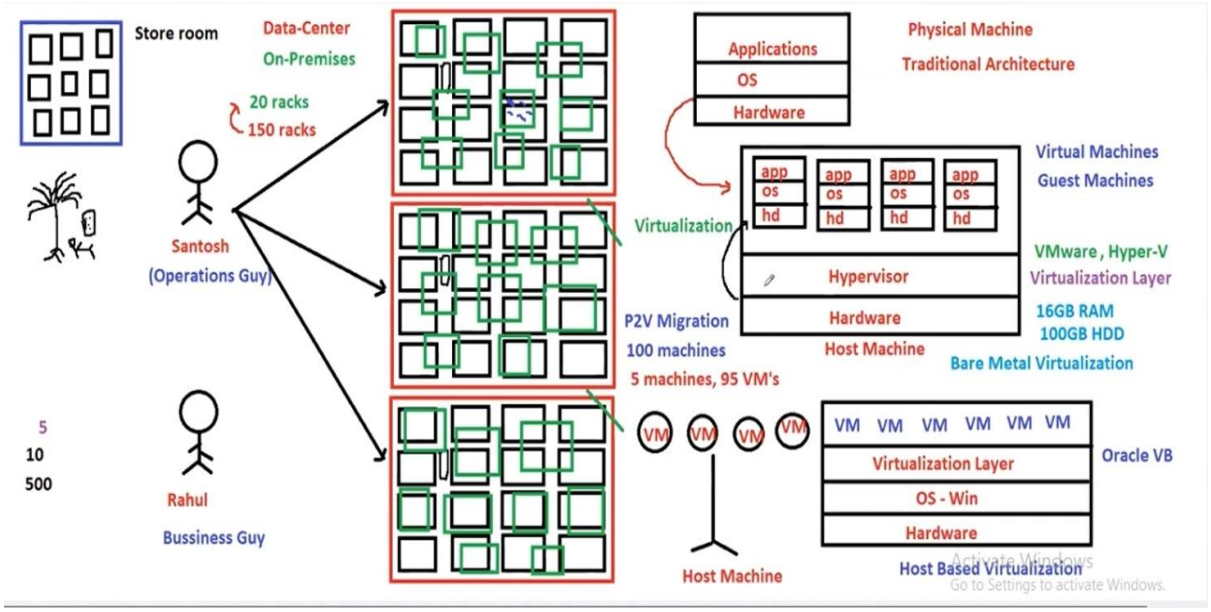
Session Layer

Transport Layer (TCP)

Network Layer (IP)

Data Link Layer

Physical Layer


Store room
Data-Center
On-Premises
20 racks
150 racks
Santosh
(Operations Guy)
Rahul
Bussiness Guy
5
10
500
Applications
OS
Hardware
Physical Machine
Traditional Architecture
app
os
hd
Virtual Machines
Guest Machines
Virtualization
Hypervisor
Hardware
Host Machine
VMware , Hyper-V
Virtualization Layer
16GB RAM
100GB HDD
Bare Metal Virtualization
P2V Migration
100 machines
5 machines, 95 VM's
VM
Virtualization Layer
OS - Win
Hardware
Oracle VB
Host Machine
Host Based Virtualization
Activate Windows
Go to Settings to activate Windows.

AWS has Global Infrastructure

AWS is providing Infrastructure as a Service

AWS has DataCenters

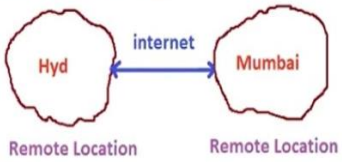
Cloud is present in the remote location

Remote Location = DataCenter

We need internet to connect to the Cloud

AWS is a Cloud Provider, who provides infrastructure as a Service

Amazon Web Services



AWS is Public Cloud Provider who provides Infrastructure as a Service

Key-Words

Virtualization, Host Machine, VM's, Infrastructure, DataCenter, Load Balancer, FireWall Protocols, Cloud, Remote Location

Physical DC --> Virtualiztion --> Cloud --> AWS (remote location(DataCenters))

Cloud Computing

Instead of doing computing on local machine / on-prem, you will be now doing computing on remote location(Cloud) that is called Cloud Computing

Deployment Models (Types of Cloud)

Public Cloud : The Provider Services which are accessed by everyone Like EX: AWS , Azure, GCP
Private Cloud : The Services which are accessed with in the organization like Ex: IBM
Hybrid Cloud : The Combination of Public and Private Cloud
Community Cloud : It is same as Private Cloud, but can be accessiable from few organizations

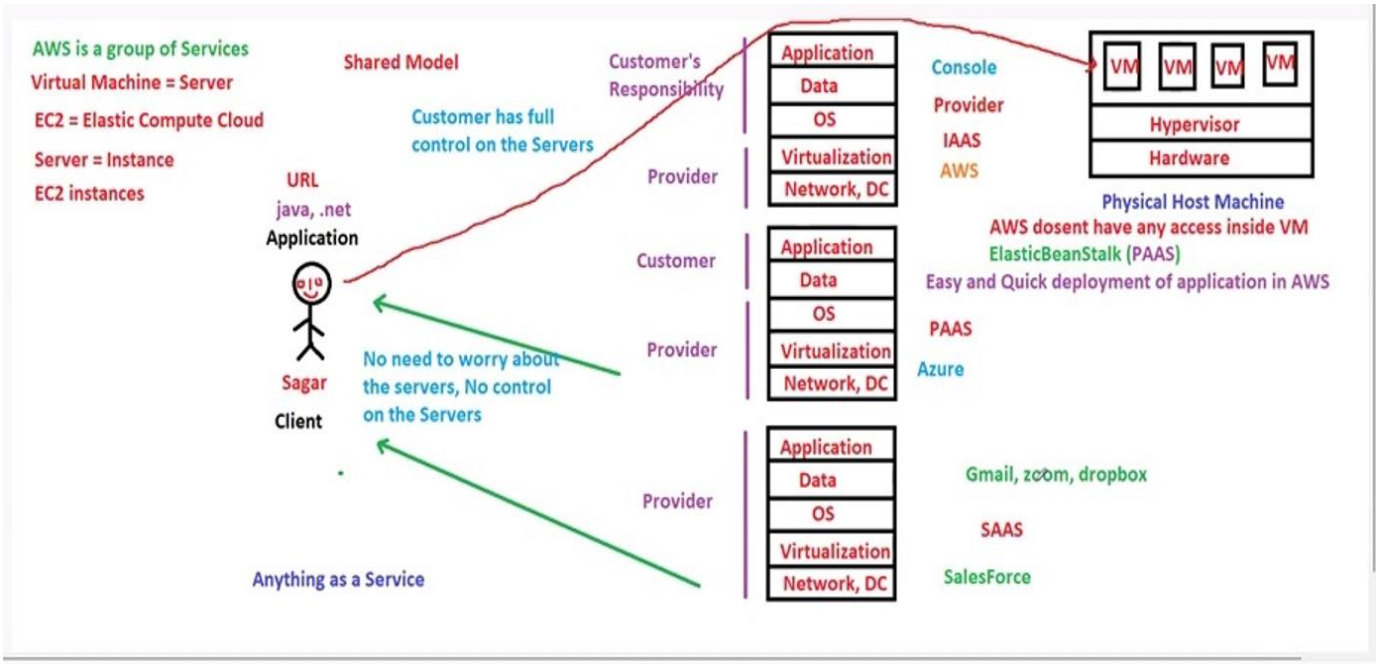
Service Models

Infrastructure as a Service (IAAS)

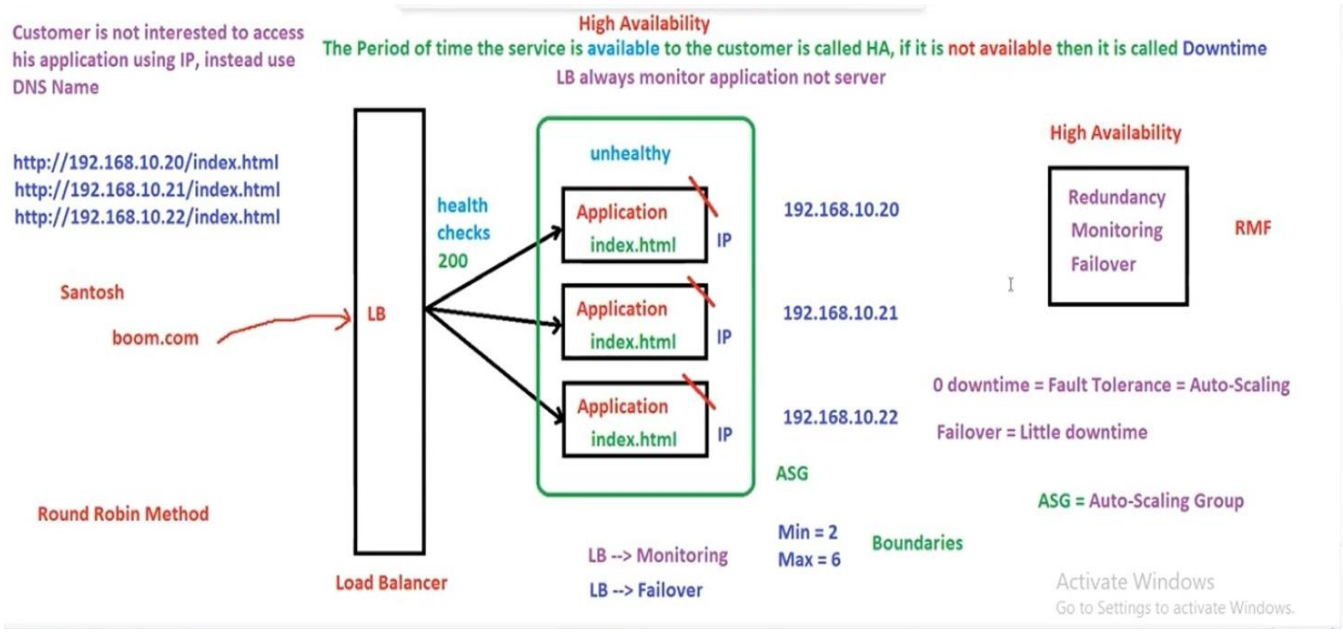
Platform as a Service (PAAS)

Software as a Service (SAAS)

Activate Windows
Go to Settings to activate Windows.




Elasticity
Increasing and decreasing the number of servers to meet increasing or decreasing work loads
Elasticity is Short term
Elasticity can be acheived in AWS using Auto-Scaling
Auto-Scaling = Scale Out and Scale In
adding
increasing
removing
decreasing
Elasticity is also called as Horizontal Scaling
LB
App
4GB RAM
App
4GB RAM
App
4GB RAM
App
4GB RAM
App
4GB RAM
Scalability
DB Server
100 DB's
5TB
32GB RAM / CPU
8GB RAM / CPU
Increasing the capacity of the server is called Scalability
Scalability = Scale Up and Scale Down
Scalability is long term
Scalability is also called as Vertical Scaling


Customer is not interested to access his application using IP, instead use DNS Name
High Availability
The Period of time the service is available to the customer is called HA, if it is not available then it is called Downtime
LB always monitor application not server
http://192.168.10.20/index.html
http://192.168.10.21/index.html
http://192.168.10.22/index.html
Santosh
boom.com
LB
health checks 200
unhealthy
Application index.html
IP
Application index.html
IP
Application index.html
IP
192.168.10.20
192.168.10.21
192.168.10.22
ASG
Round Robin Method
Load Balancer
LB --> Monitoring
LB --> Failover
Min = 2
Max = 6
Boundaries
High Availability
Redundancy
Monitoring
Failover
RMF
0 downtime = Fault Tolerance = Auto-Scaling
Failover = Little downtime
ASG = Auto-Scaling Group
Activate Windows
Go to Settings to activate Windows.

AWS has Global Infrastructure

Servers = Instances

Region is a place where AWS has its own infrastructure

A Region has multiple DataCenter

A Region has multiple AZ's

Servers/Instances are placed in AZ's

AZ's are in sync with each other(network), but not data

Best practise is to distribute the instances across multiple AZ's

Very very less chance that 1 AZ goes down

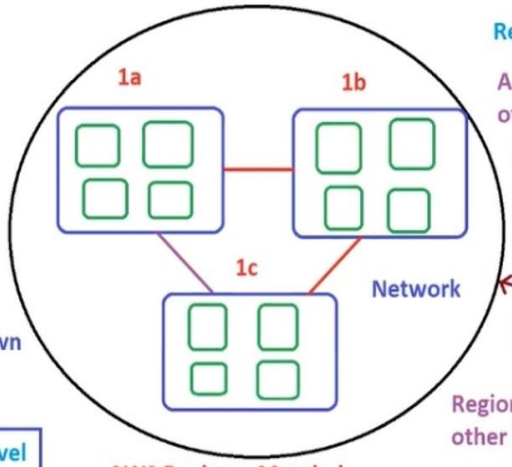
AZ = group of DC's

1a	1b
andheri,bandra,dadar	thane, kurla,panvel

Regions and Availability Zone

Region = Its a geo-graphical area , Ex : AWS-Region =Mumbai

Availability Zone = Its a simply a DataCenter (AZ)



AWS Region = Mumbai

Mumbai = ap-south-1

AZ's = ap-south-1a

= ap-south-1b

= ap-south-1c

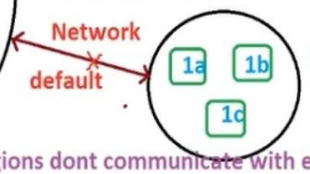
Regions and AZ's are managed by AWS

AZ's can communicate with each other by default

AZ's networks are inter-connected

Low Latency = Good

High Latency = Bad



Regions dont communicate with each other by default, if required yes

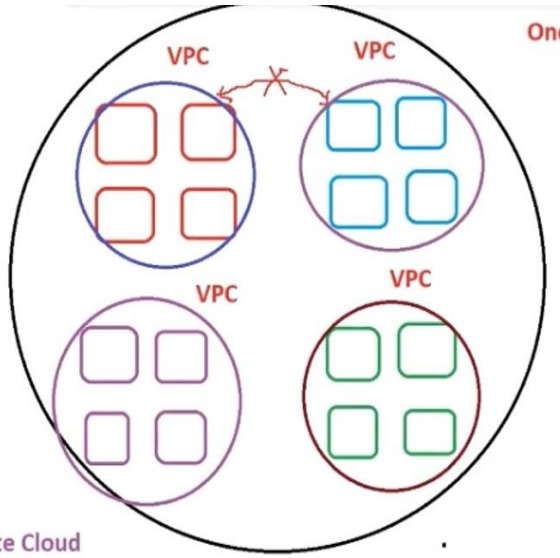
Activate Windows
Go to Settings to activate Wi

Mahesh

Srinu

Santosh

Sagar

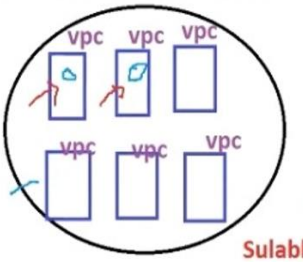


Virtual Private Cloud

VPC

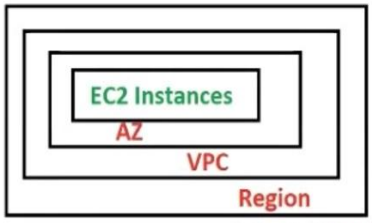
AWS

One Region can have multiple VPC's (Max 5 VPC's)



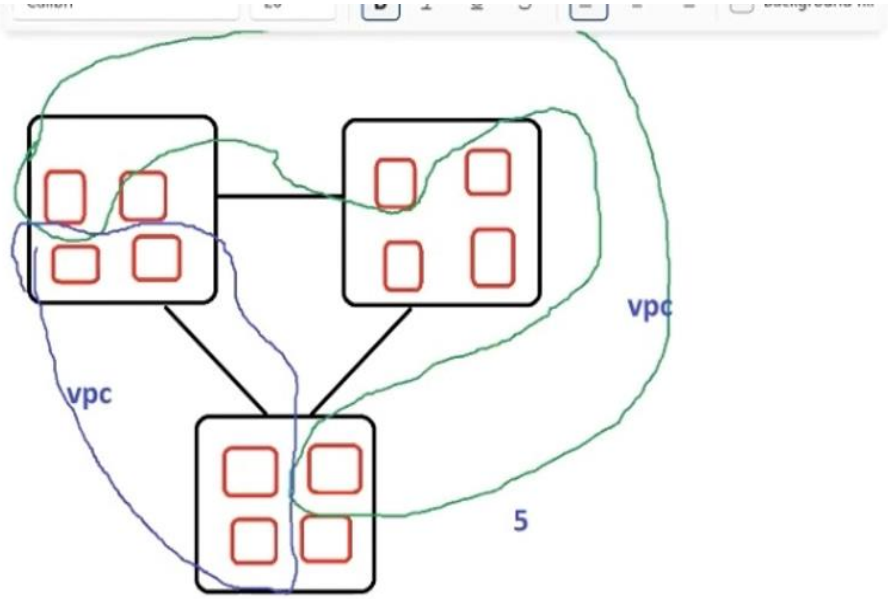
AWS

Sulabh Complex



AWS

Activate Windows
Go to Settings to activate W



VPC spans across AZ's

Load Balancer distributes the traffic to multiple EC2 instances across multiple AZ's

AWS Services can be
Regional or Global

EC2 is Regional

EC2 = Elastic Compute Cloud
In EC2 Service, We can launch EC2 instances
Servers = Instances / EC2 Instances (VM)
Load Balancer = Which distribute the traffic to the Servers
Elastic Load Balancer(ELB) = ELB distribute the traffic to multiple EC2 instances across multiple AZ's
ELB is completely managed by AWS (HA, AS, Scalability, Performance etc)
ELB is not a Server, it is a Service for us
You cannot login to ELB, you can access ELB with the DNS Name
ELB dosent have AZ's. It is created regional level

lasticBeanStalk = Easy and Quick deployment of applications in AWS
n General, in PAAS --> You dont have any control on the Servers
n AWS BeanStalk --> You have full control on the EC2 instances
aunched by BeanStalk
eanStalk handles EC2 instances(OS) behalf of us

ightSail = If you want to setup and create a virtual private
erver which already has everything installed(Wordpress,
itlab, node.js, joomla, drupal, Django, Ghost, Nginx,
edmine etc) --> No Auto-Scaling

App
Client
java

EC2: Launch EC2 instances
Configure and deploy application

BeanStalk
Just Select the Platform

Tomcat
.NET
Docker
Python
GO

App
EC2

HA
Auto-Scaling
Scalability

LB → App → DB

On-Premises

DNS Name

Scale OUT
Scale IN

ELB

EC2 1a
EC2 1b
EC2 1c

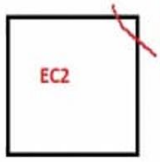
Lambda is used for automation

Lambda

Lambda is Serverless

Create Function

java, python, ruby etc



Terminated

Launch Event

triggered

invoke

Python

Code

to terminate the EC2 instance which has been launched

Lambda is invoked based on the triggers

Launch
Stop
Start
Reboot
Terminate --> Destroy
Kill



Schedule

Lambda

9PM ---> STOP -----> StopFunction (Stop all EC2 instances)

9AM --> START ----> Start Function (Start all EC2 instances)

Stop or Start EC2 instances automatically based on the time

In AWS, all services will start with Simple and end with Service

SNS --> Simple Notification Service
SES --> Simple Email Service

S3 -Simple Storage Service

S3 is unlimited Storage by AWS

S3 is used to just store your files

S3 can store all FLAT files

With S3, you can upload, download and access your files

You cannot execute any files in S3

Floppy --> 2MB
CD's --> 700MB
DVD's --> 4.7GB
Pen Drives --> 128GB
Hard Disks --> 2TB

S3 is a Object Based Storage

Is it possible to install OS in S3? NO
Is it possible to install DB in S3? NO
Is it possible to run .net, .py etc in S3 ? NO

S3 is Global

Buckets are regional

S3 is Serverless

AWS handles HA, Performance, Scalability etc for S3

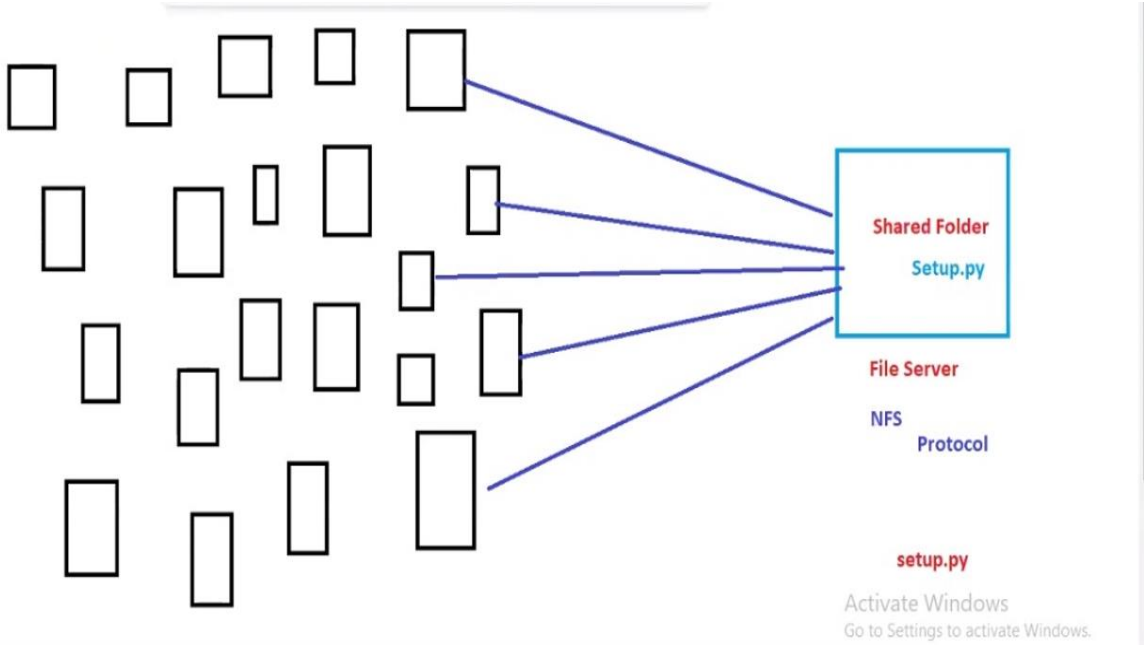
Windows	S3
Folder	Bucket
Files	Objects
boom.mp4	KEY

Bucket is a container for Objects

Object is a File

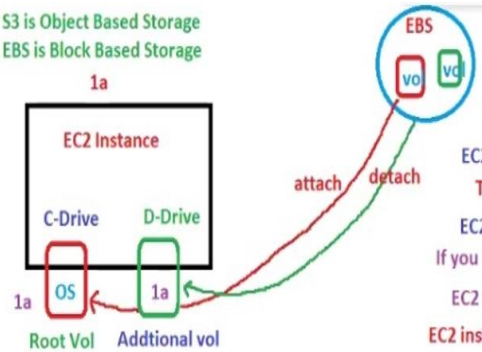
Key is the File name / Name of the Object

S3 supports STATIC website hosting (HTML files)[Create a bucket, upload all files, enable static website hosting]
(No need to worry about HA, performance, etc handled by S3)



Activate Windows
Go to Settings to activate Windows.

S3 is Object Based Storage
EBS is Block Based Storage



Root Vol Addtional vol

Device Name

/dev/sda1 /dev/sdb, e, f, g etc

/dev/xvda xvdf, xvdg , etc

Root Volume is always mounted / attached as /dev/sda1

EBS - Elastic Block Storage

Hard Disk = Volume = EBS Volume

Volumes can be attached and detached

You can attach multiple volumes to the EC2 instance

EC2 instance has a default volume and that is called ROOT Volume

The Root Volume always contain OS (Win, Linux , Mac)

EC2 supports only Server side OS not Client side OS

If you have OS on the Volume, that volume is called ROOT Volume

EC2 instance can have only 1 ROOT Volume

EC2 instance can have multiple addtional Volumes

Max size of each EBS Volume is 16TB

You cannot attach a volume to multiple EC2 instances at the same time

Volumes should be pre-provisioned like 50GB, 100GB ... max 16TB

Volume size can be increased on FLY (no need to stop the EC2 instance)

Volume size cannot be decreased

Is it possible to detach the ROOT Volume while EC2 is running? NO: Stop the EC2 instance first and then detach the ROOT Volume

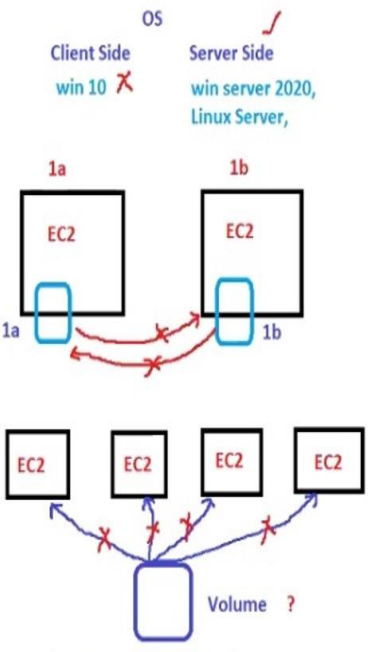
Is it possible to detach ADDTIONAL Volume while EC2 is running? YES: it is not recommended to detach while running, stop first

EC2 instance has AZ, Volume also have AZ

EC2 instance and Volume should be in same AZ

We cannot attach 1a volume to 1b EC2 instance (diff AZ)

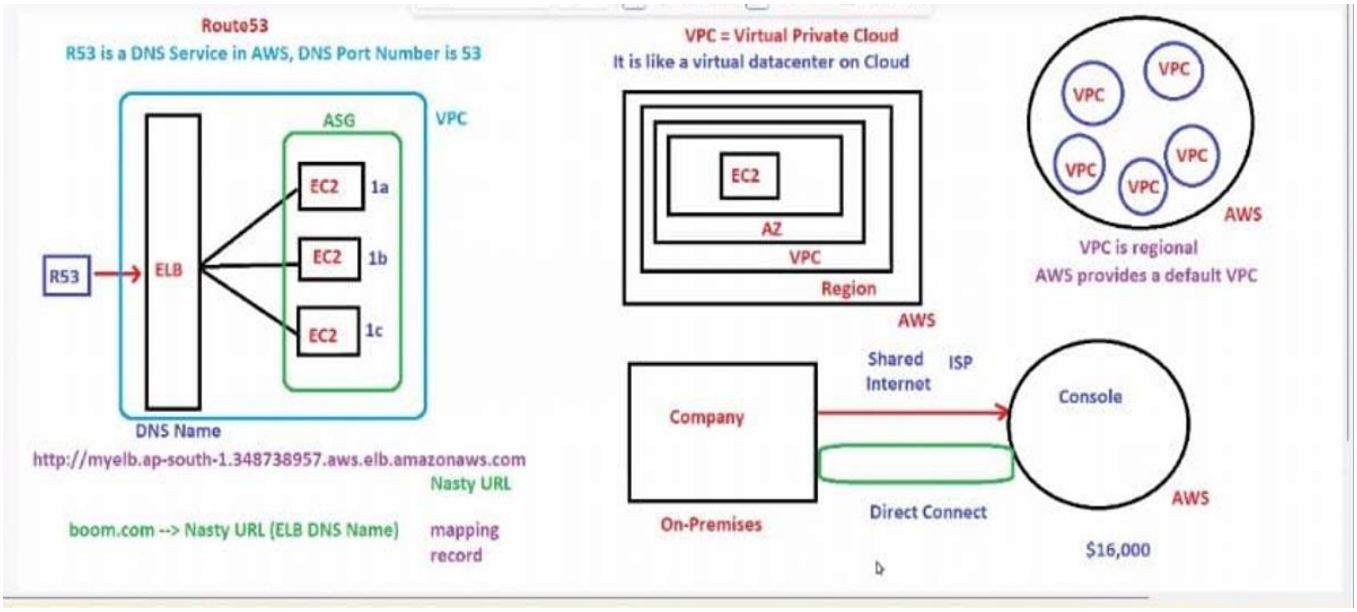
We can attach 1a volume to 1a EC2 instance (same AZ)



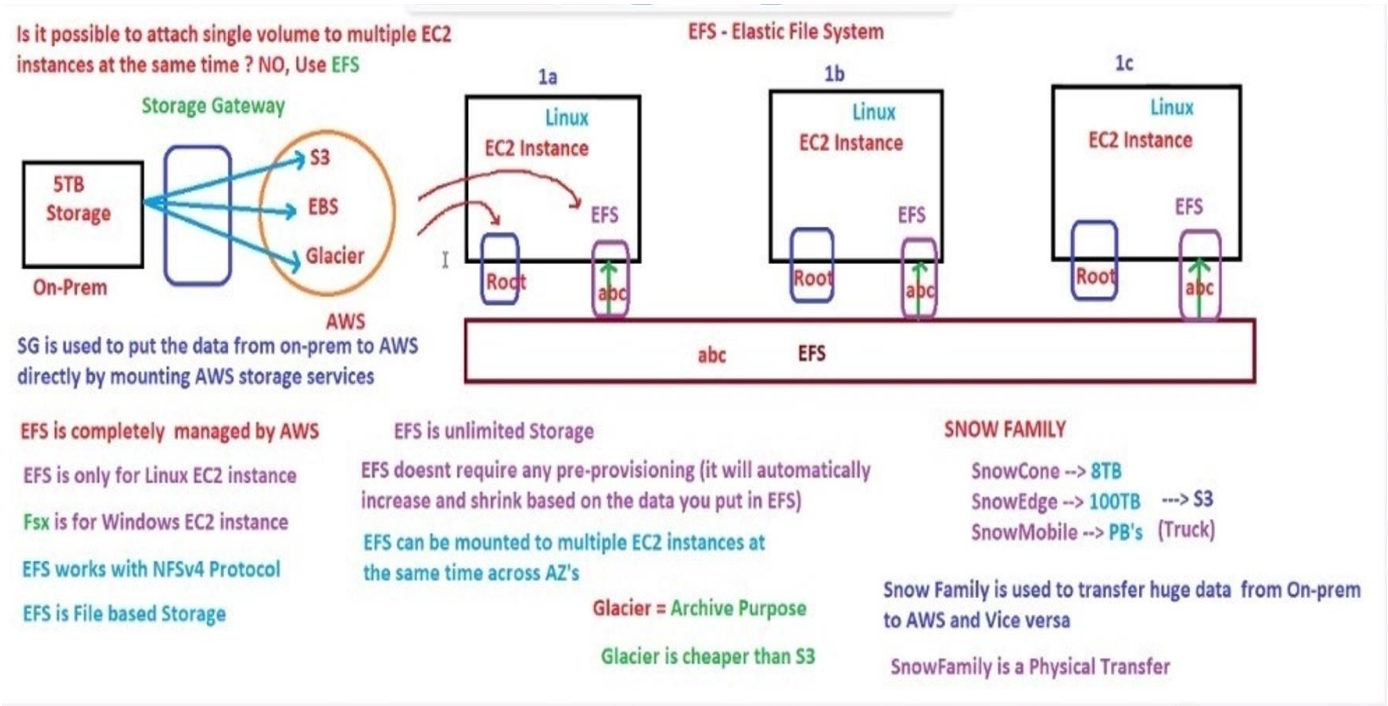
Volumes cannot be shared

Activate Windows
Go to Settings to activate Windows.

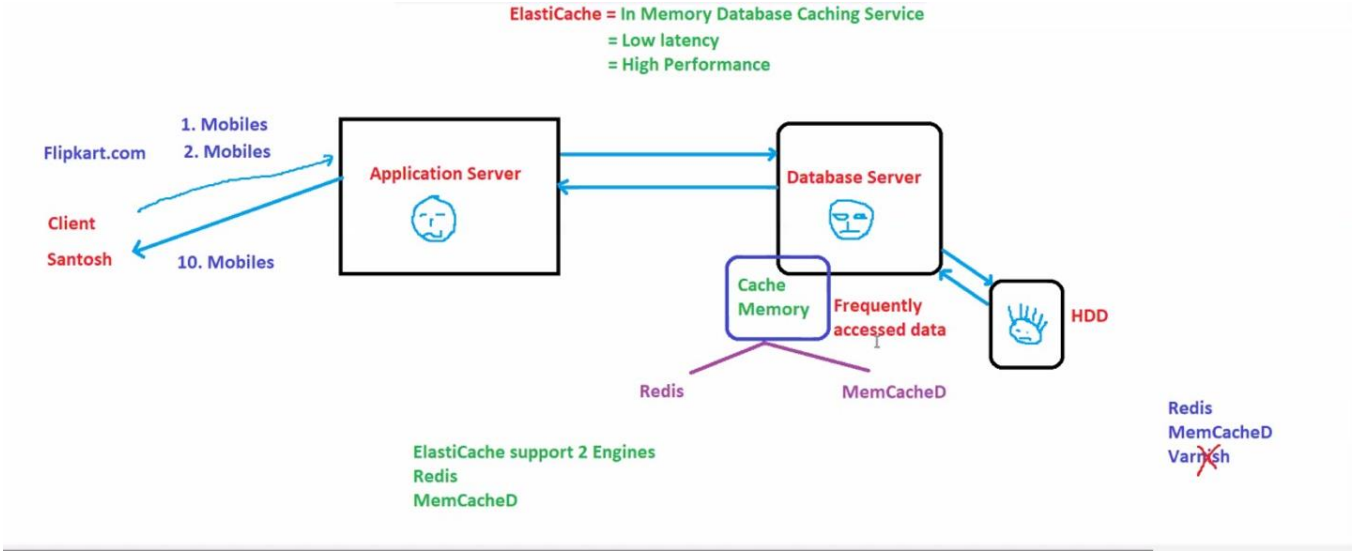
Route 53



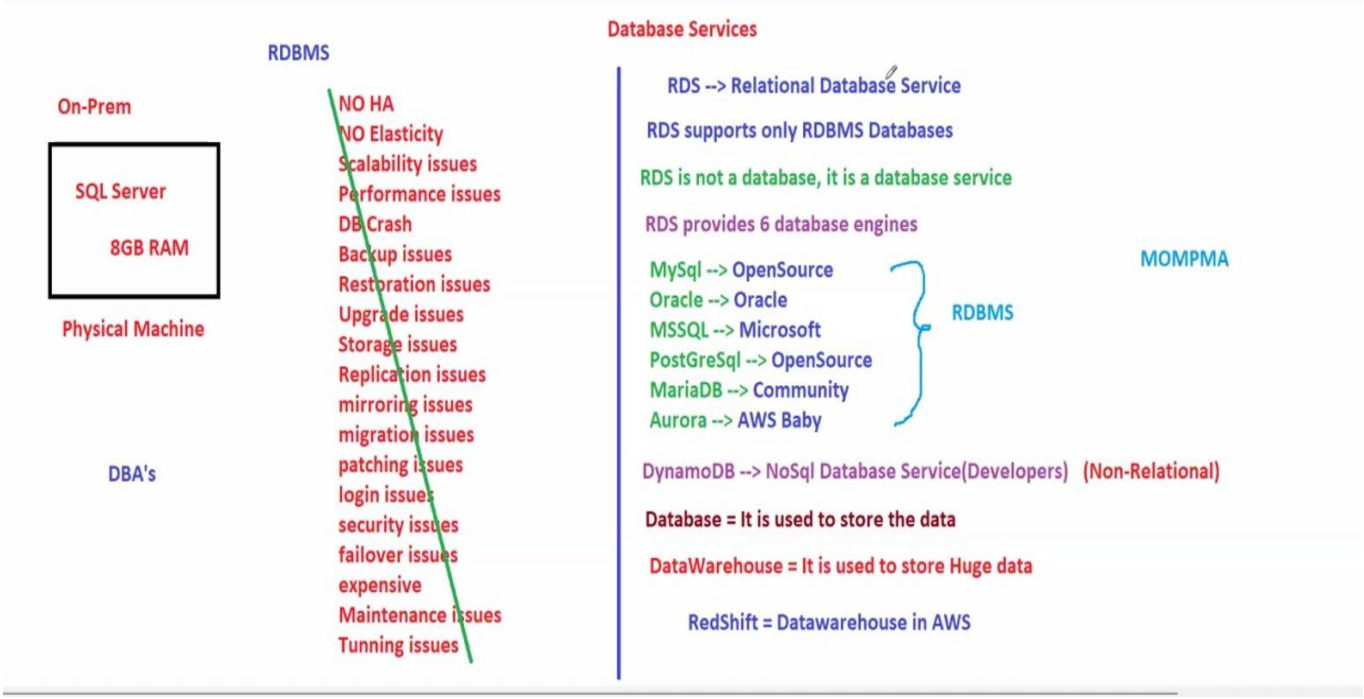
Elastic File System



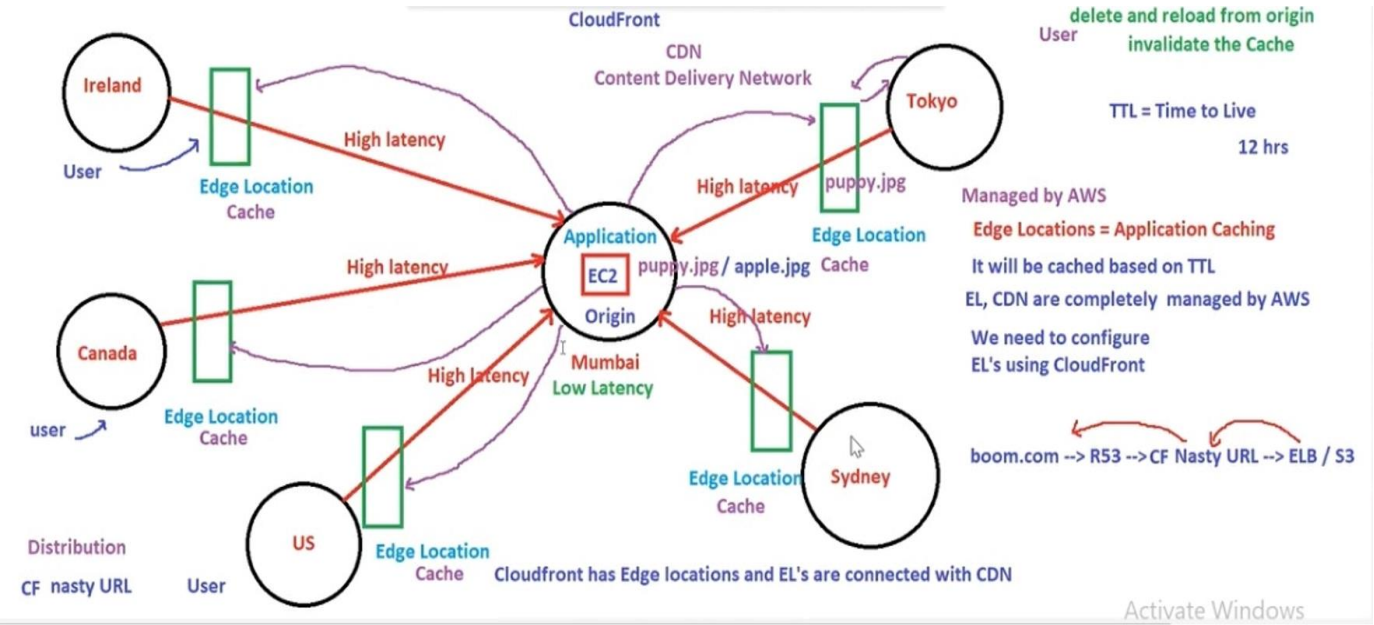
ElastiChache



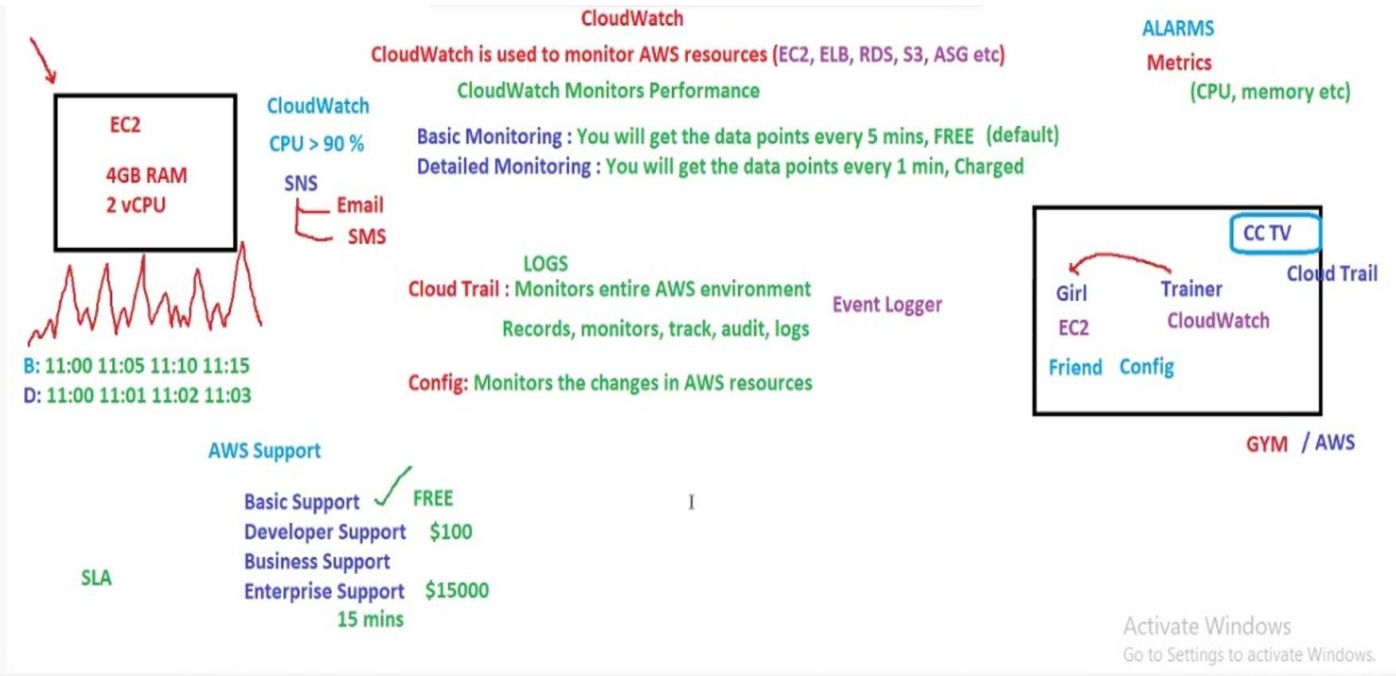
Relational Database Service :



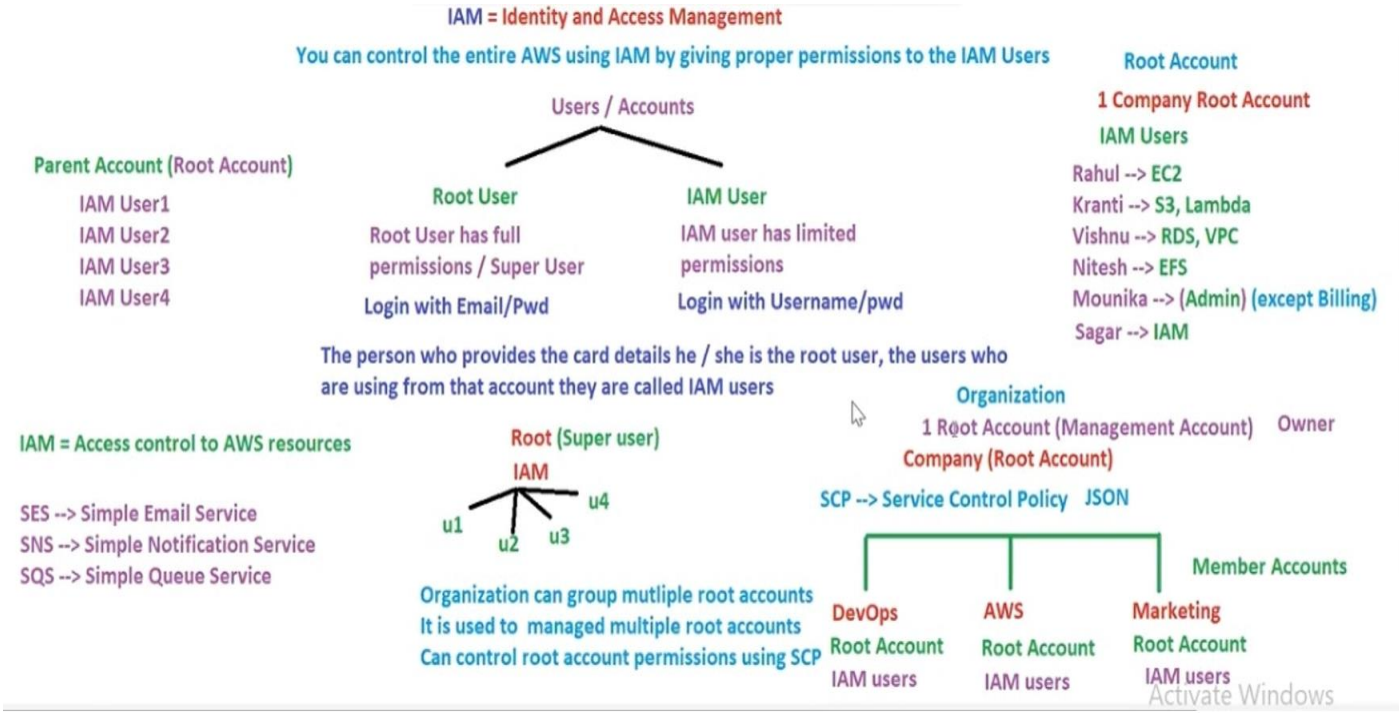
CloudFront :



CloudWatch :



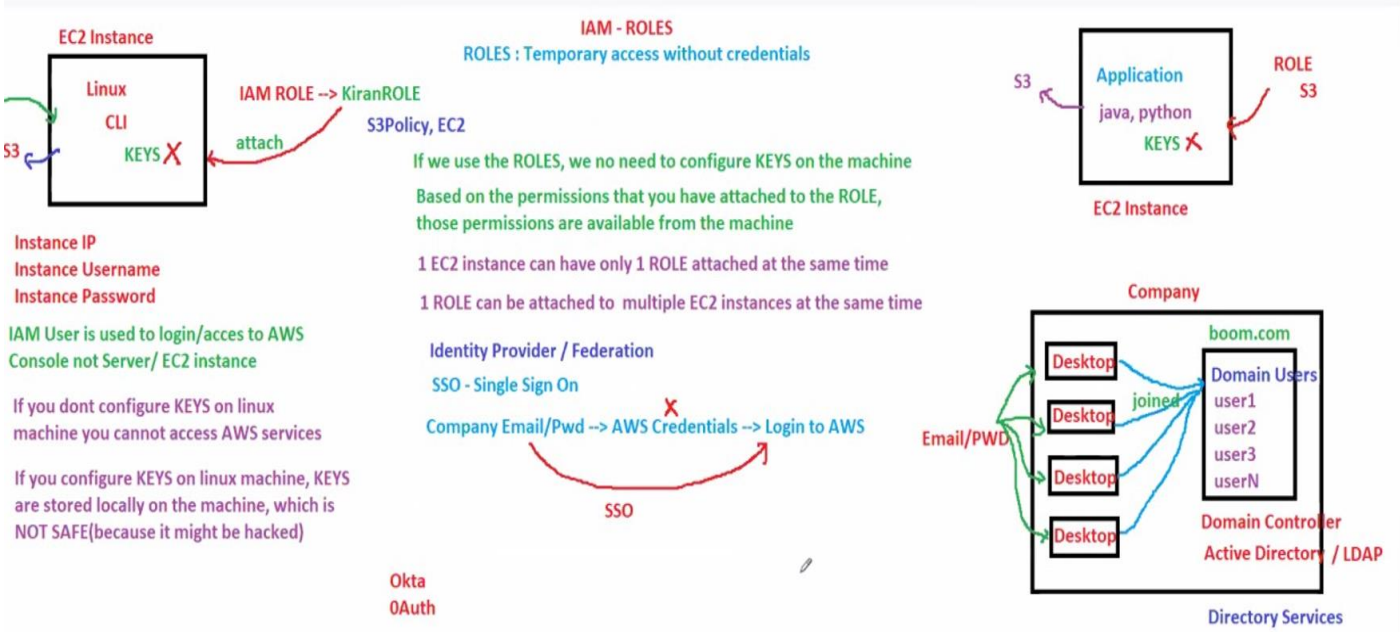
IAM

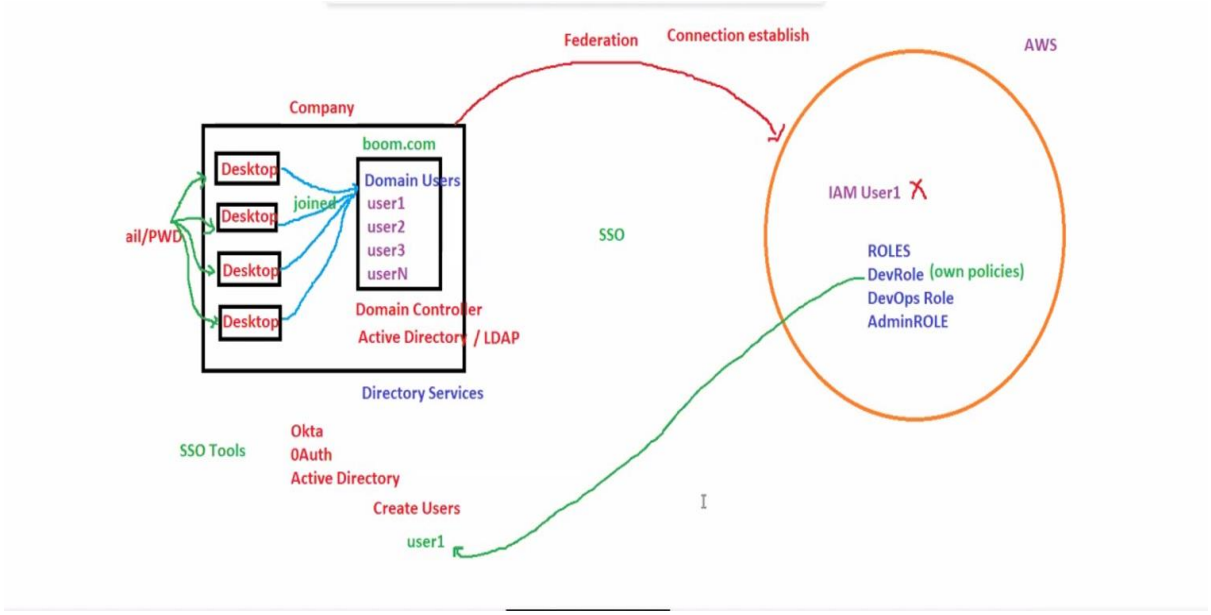


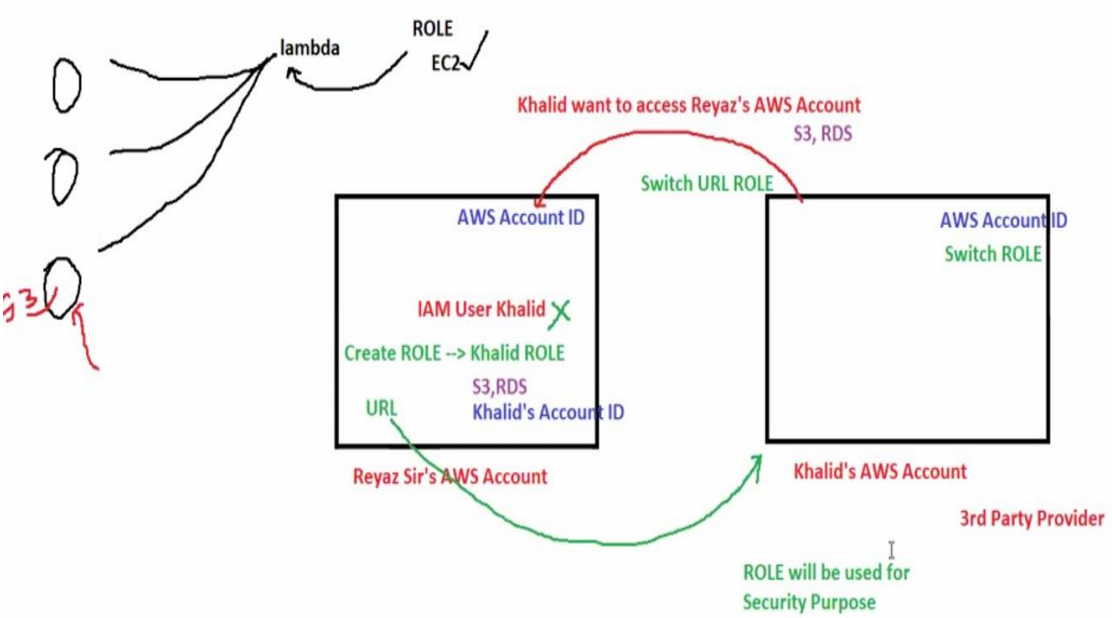
IAM Groups



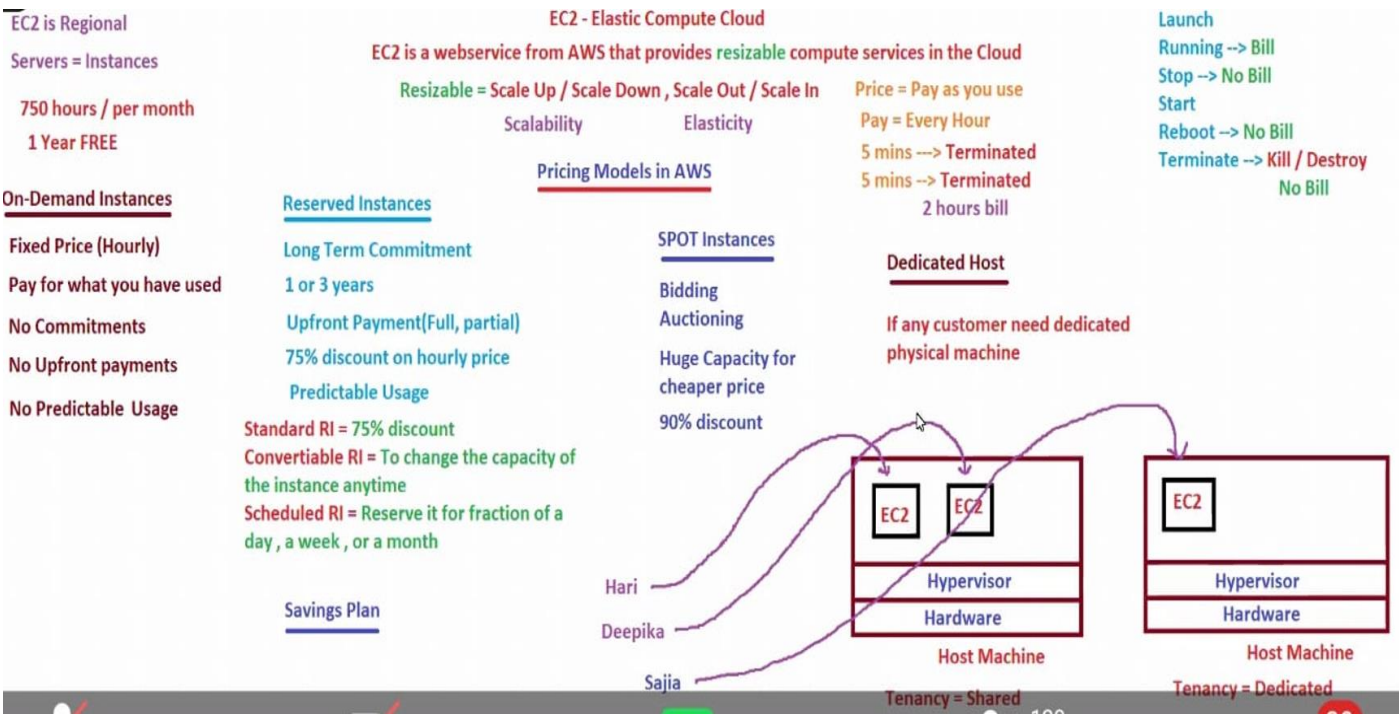
IAM-ROLES


EC2 Instance
Linux
CLI
KEYS
S3
IAM ROLE --> KiranROLE
attach
S3Policy, EC2
IAM - ROLES
ROLES : Temporary access without credentials
If we use the ROLES, we no need to configure KEYS on the machine
Based on the permissions that you have attached to the ROLE, those permissions are available from the machine
1 EC2 instance can have only 1 ROLE attached at the same time
1 ROLE can be attached to multiple EC2 instances at the same time
Application
java, python
KEYS
ROLE
S3
EC2 Instance
Instance IP
Instance Username
Instance Password
IAM User is used to login/acces to AWS Console not Server/ EC2 instance
If you dont configure KEYS on linux machine you cannot access AWS services
If you configure KEYS on linux machine, KEYS are stored locally on the machine, which is NOT SAFE(because it might be hacked)
Identity Provider / Federation
SSO - Single Sign On
Company Email/Pwd --> AWS Credentials --> Login to AWS
SSO
Okta
0Auth
Company
boom.com
Desktop
Desktop
Desktop
Desktop
joined
Email/PWD
Domain Users
user1
user2
user3
userN
Domain Controller
Active Directory / LDAP
Directory Services

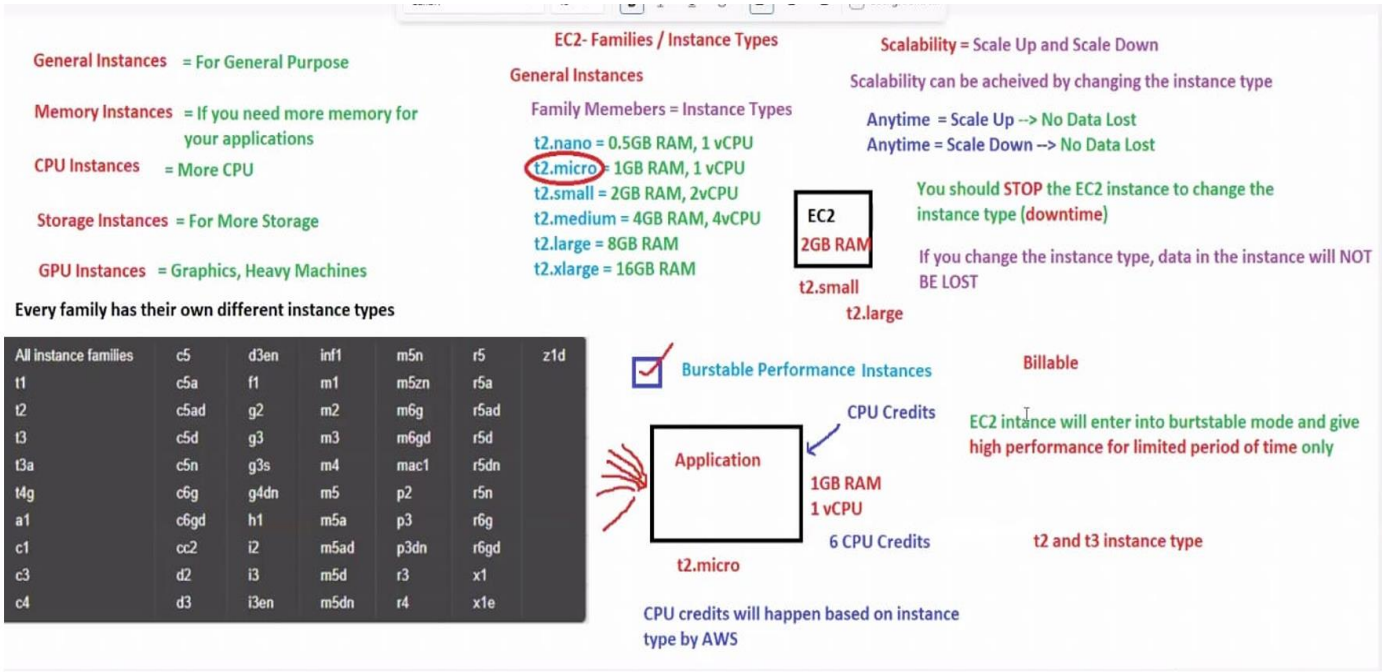

Federation
Connection establish
AWS
Company
boom.com
Desktop
Desktop
Desktop
Desktop
joined
ail/PWD
Domain Users
user1
user2
user3
userN
Domain Controller
Active Directory / LDAP
Directory Services
SSO
IAM User1
ROLES
DevRole (own policies)
DevOps Role
AdminROLE
SSO Tools
Okta
0Auth
Active Directory
Create Users
user1



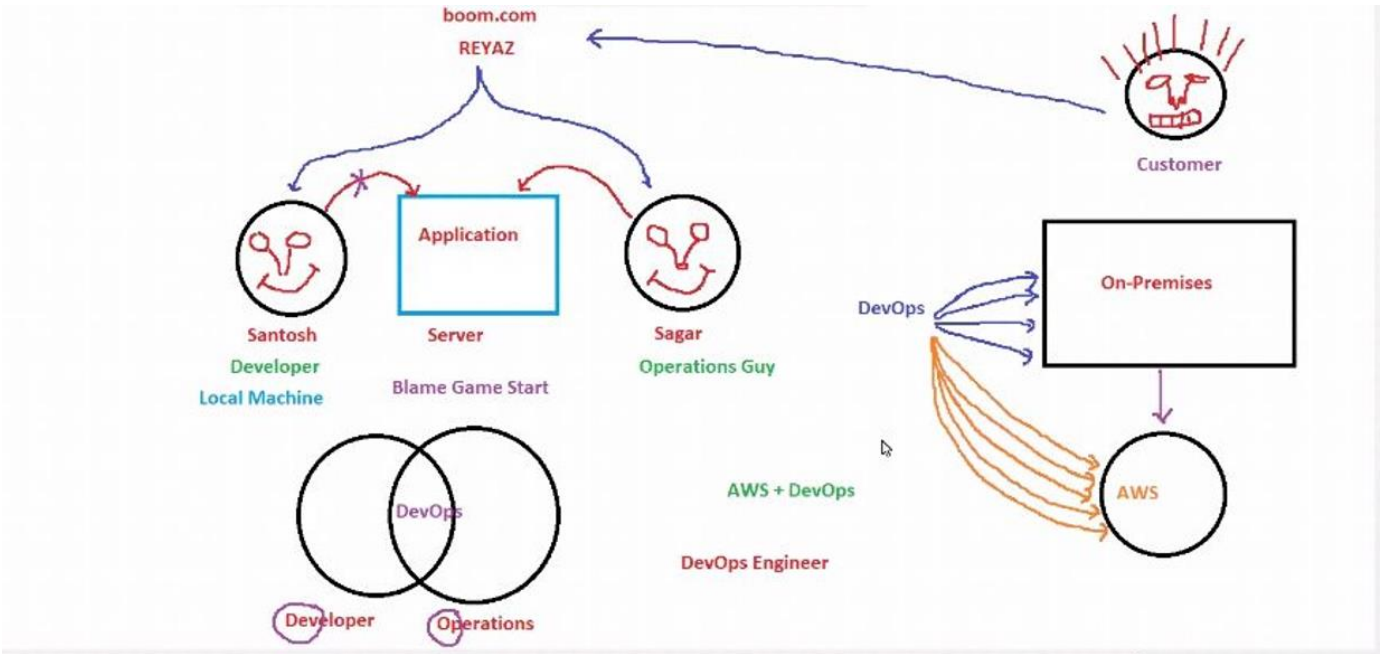
Elastic Compute Cloud

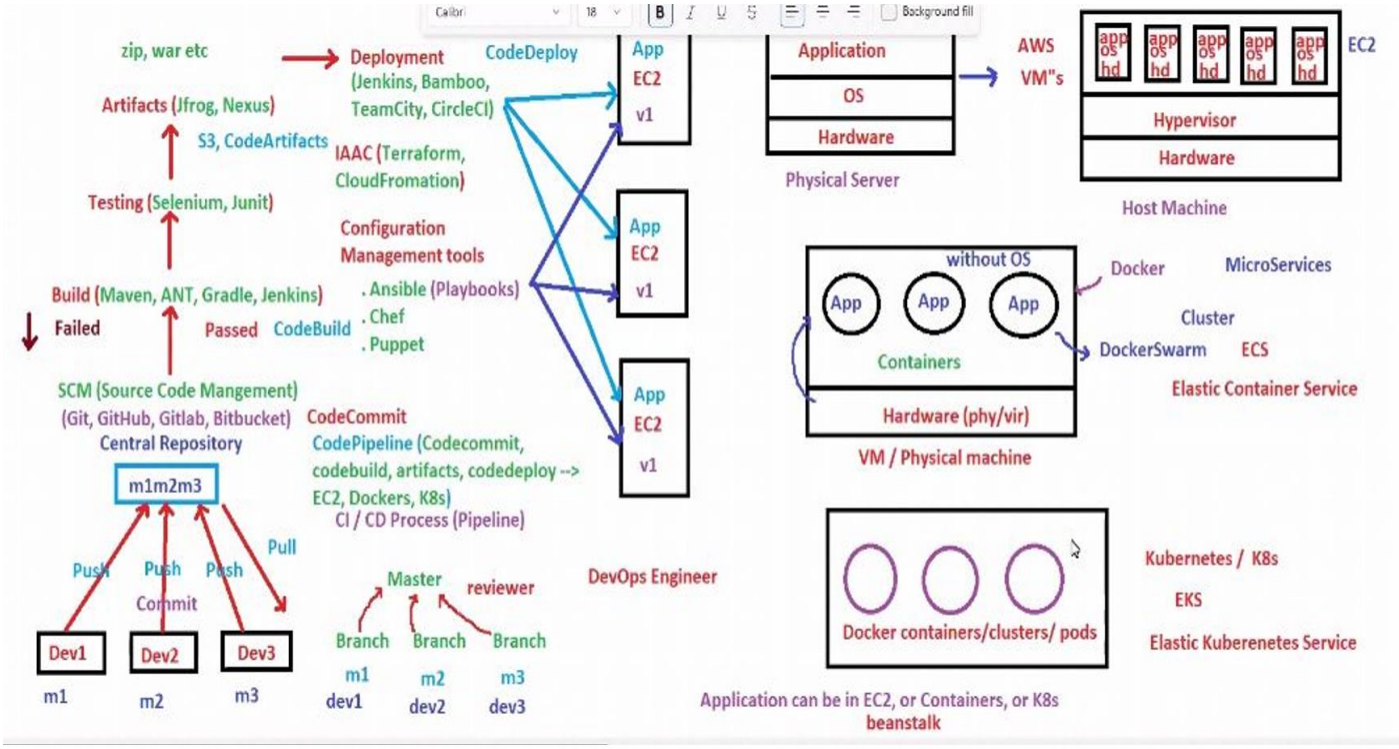


EC2-Families

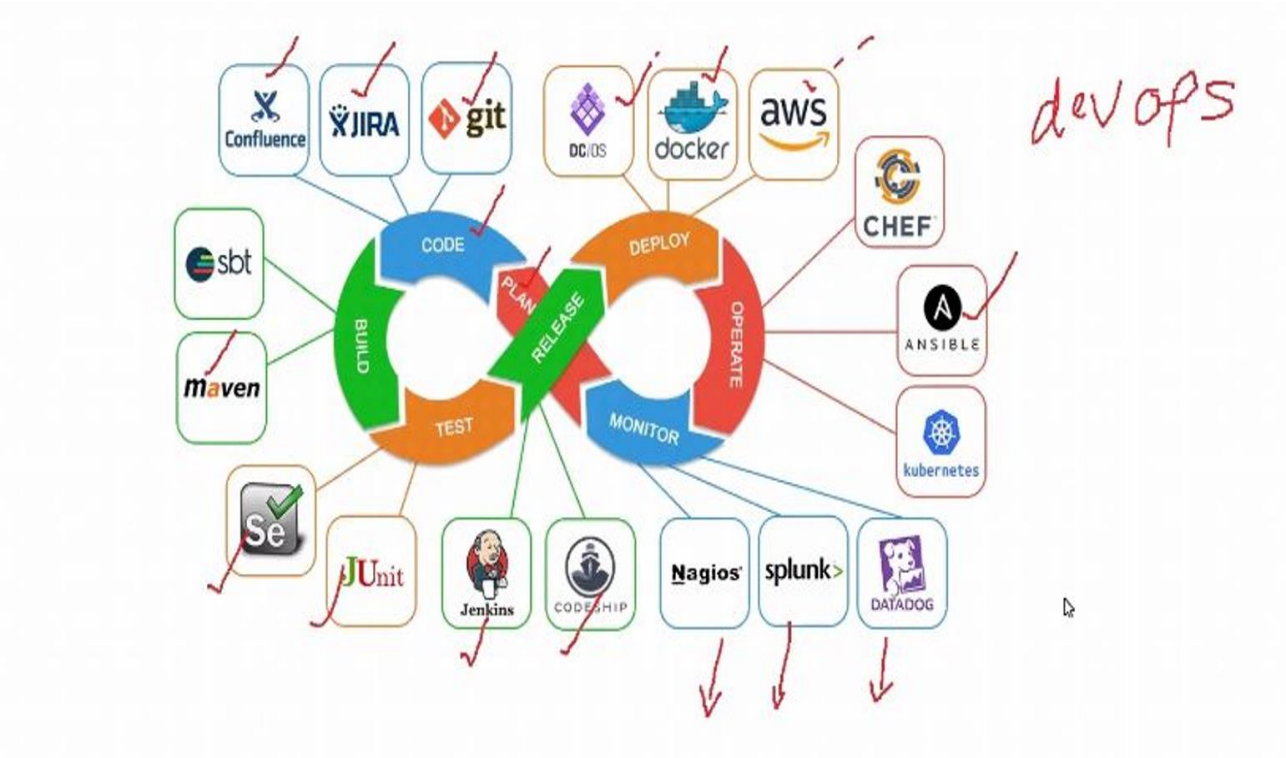

General Instances = For General Purpose
Memory Instances = If you need more memory for your applications
CPU Instances = More CPU
Storage Instances = For More Storage
GPU Instances = Graphics, Heavy Machines
Every family has their own different instance types
EC2- Families / Instance Types
General Instances
Family Memebers = Instance Types
t2.nano = 0.5GB RAM, 1 vCPU
t2.micro = 1GB RAM, 1 vCPU
t2.small = 2GB RAM, 2vCPU
t2.medium = 4GB RAM, 4vCPU
t2.large = 8GB RAM
t2.xlarge = 16GB RAM
EC2
2GB RAM
t2.small
t2.large
Scalability = Scale Up and Scale Down
Scalability can be acheived by changing the instance type
Anytime = Scale Up --> No Data Lost
Anytime = Scale Down --> No Data Lost
You should STOP the EC2 instance to change the instance type (downtime)
If you change the instance type, data in the instance will NOT BE LOST
All instance families
t1
t2
t3
t3a
t4g
a1
c1
c3
c4
c5
c5a
c5ad
c5d
c5n
c6g
c6gd
cc2
d2
d3
d3en
f1
g2
g3
g3s
g4dn
h1
i2
i3
i3en
inf1
m1
m2
m3
m4
m5
m5a
m5ad
m5d
m5dn
m5n
m5zn
m6g
m6gd
mac1
p2
p3
p3dn
r3
r4
r5
r5a
r5ad
r5d
r5dn
r5n
r6g
r6gd
x1
x1e
z1d
Burstable Performance Instances
Billable
CPU Credits
Application
1GB RAM
1 vCPU
6 CPU Credits
t2.micro
EC2 intance will enter into burtstable mode and give high performance for limited period of time only
t2 and t3 instance type
CPU credits will happen based on instance type by AWS

DevOps

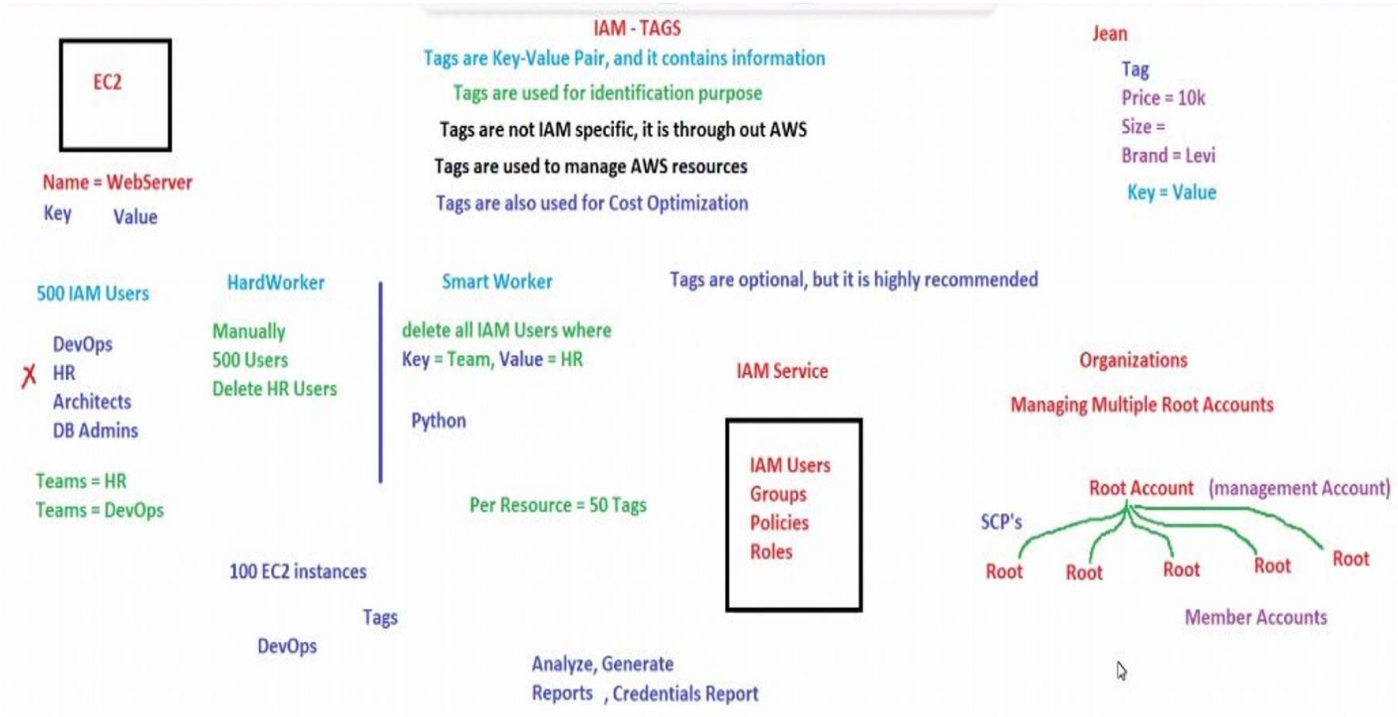

boom.com
REYAZ
Customer
Application
Server
Santosh
Developer
Local Machine
Blame Game Start
Sagar
Operations Guy
DevOps
On-Premises
AWS
AWS + DevOps
DevOps Engineer
DevOps
Developer
Operations



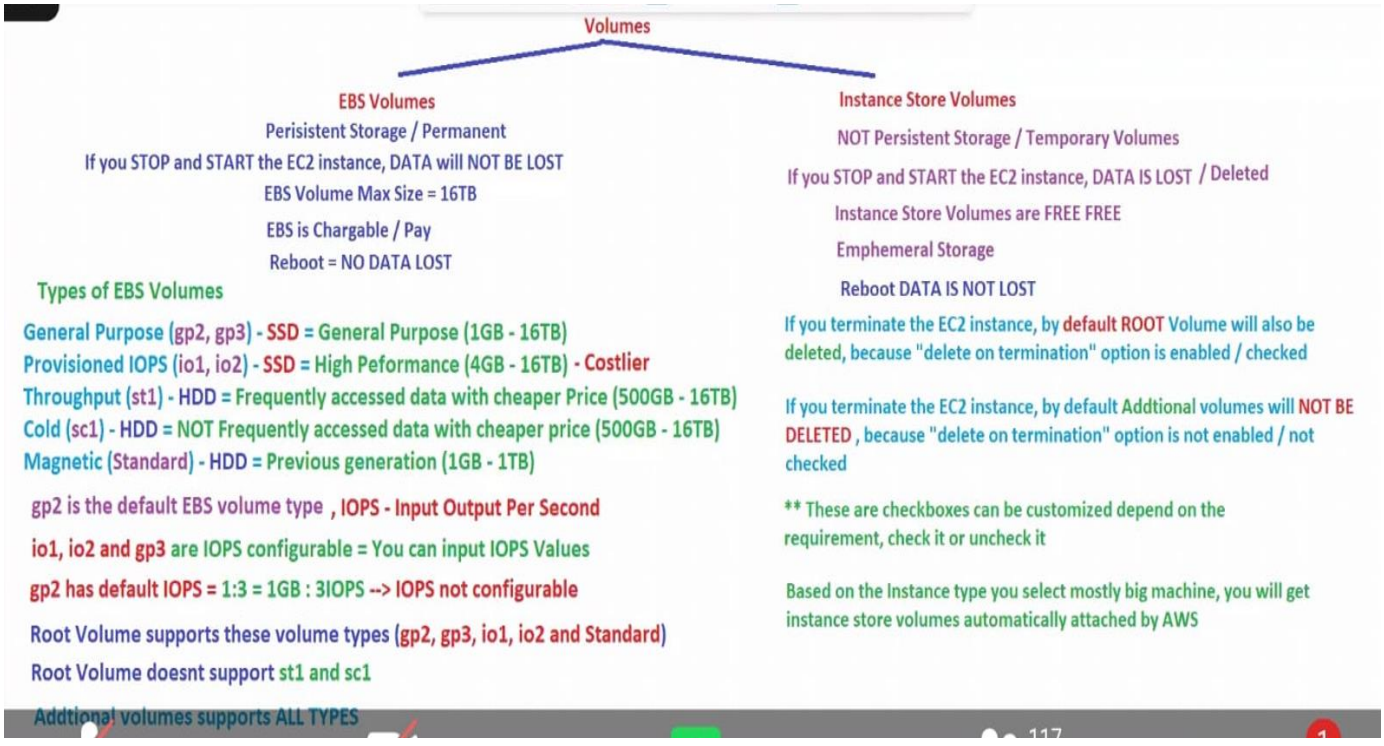
DevOps Tools

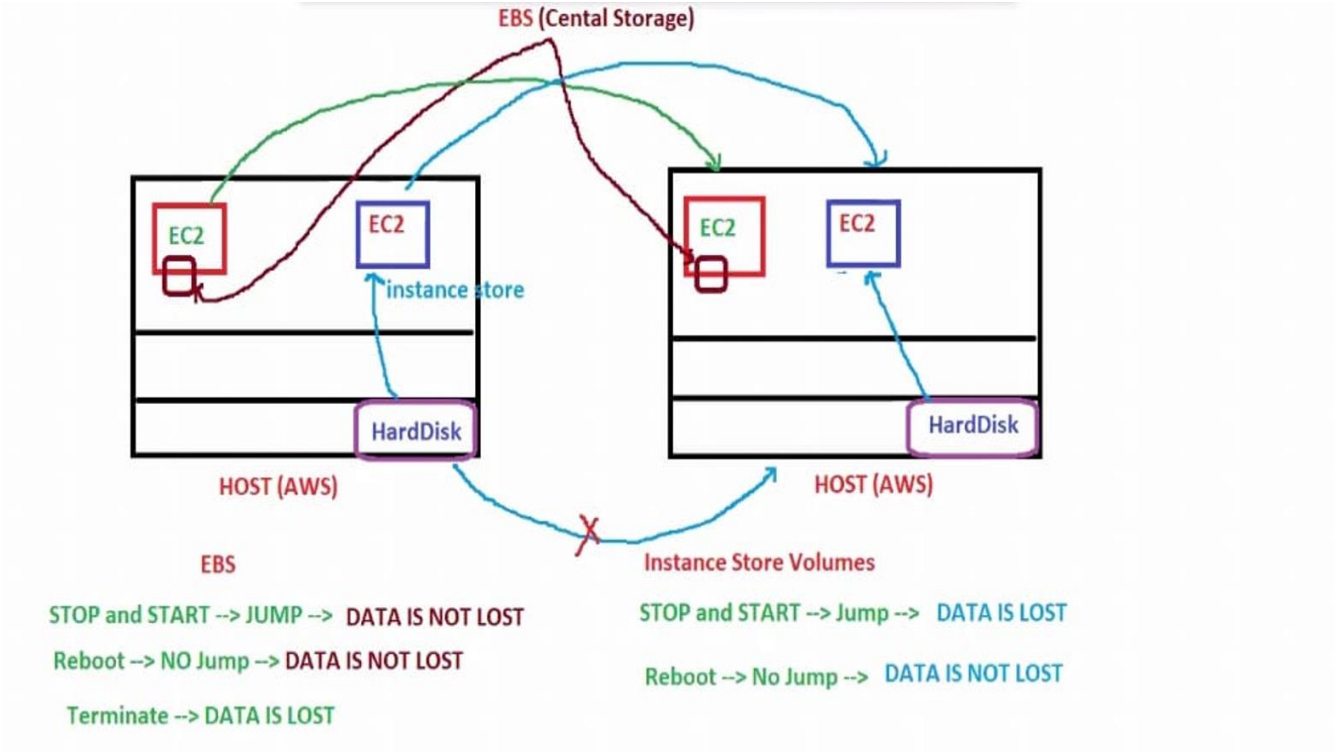


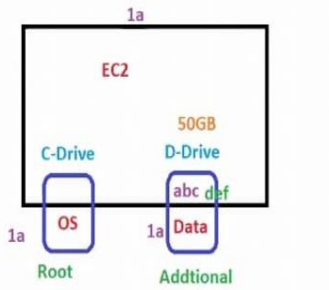
IAM Tags



Volumes







Vol 50GB --> Snapshot(50GB) --> abc

--> Snapshot (50GB) --> abc + def

--> Snapshot --> def

Incremental backup

SNAPSHOTS

1. Snapshot is a point in time copy of the volume
2. Backup of the volume is also called as Snapshot
3. EBS Snapshots are created from EBS Volumes
4. You can create a Snapshot from the Volumes
5. EBS Volume --> EBS SNAPSHOT --> EBS VOLUME
6. You cannot attach snapshots directly to the EC2 instance, you have to create a volume out of snapshot first and then attach volume to the EC2 instance
7. Is it possible to login or use snapshots directly ? NO
8. Snapshots are stored in S3 (Provider's S3)
9. Snapshots doesnt have any AZ"s
10. Snapshots are Regional
11. By default, Snapshots are Private, if required you can make it public
12. You can copy the snapshots from one region to another region in the same account
13. Snapshots can be shared from one AWS account to another AWS account using AWS account ID (Private)
14. EBS Volumes cannot be moved directly to any other region, use Snapshots

EBS Volumes are created from EBS Snapshots

Instance Store volumes are created from a template stored in S3

To create a snapshot we no need to stop the EC2 instance

By default, Volumes, snapshots are not Encrypted

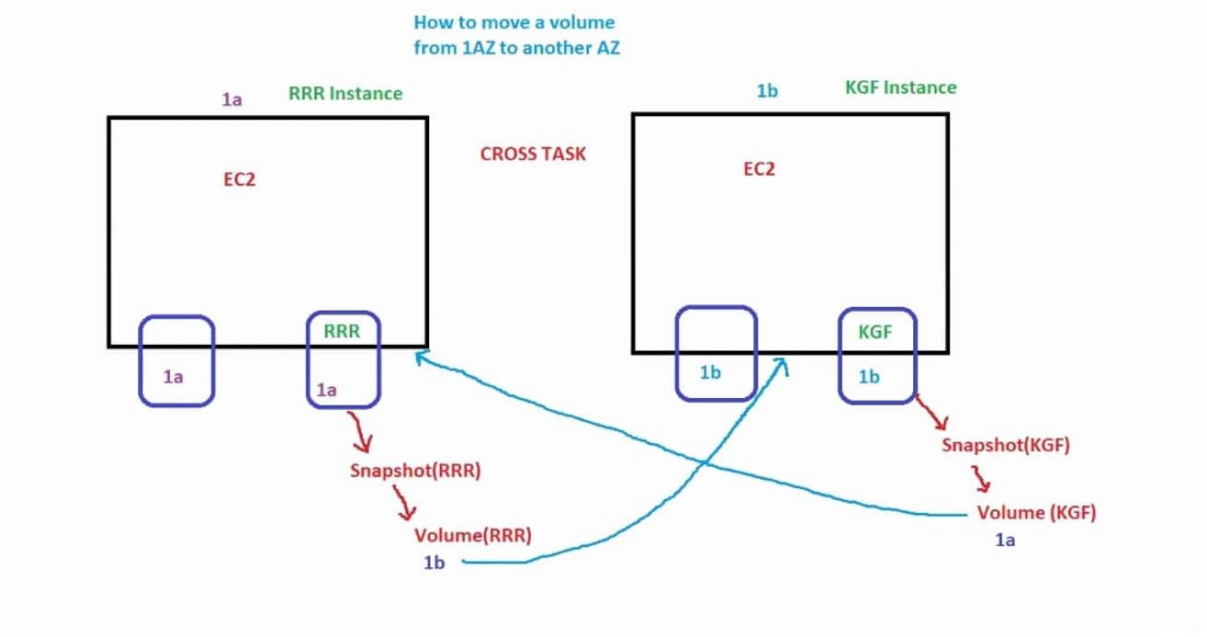
NOT Encrypted --> NOT Encrypted

Encrypted --> Encrypted

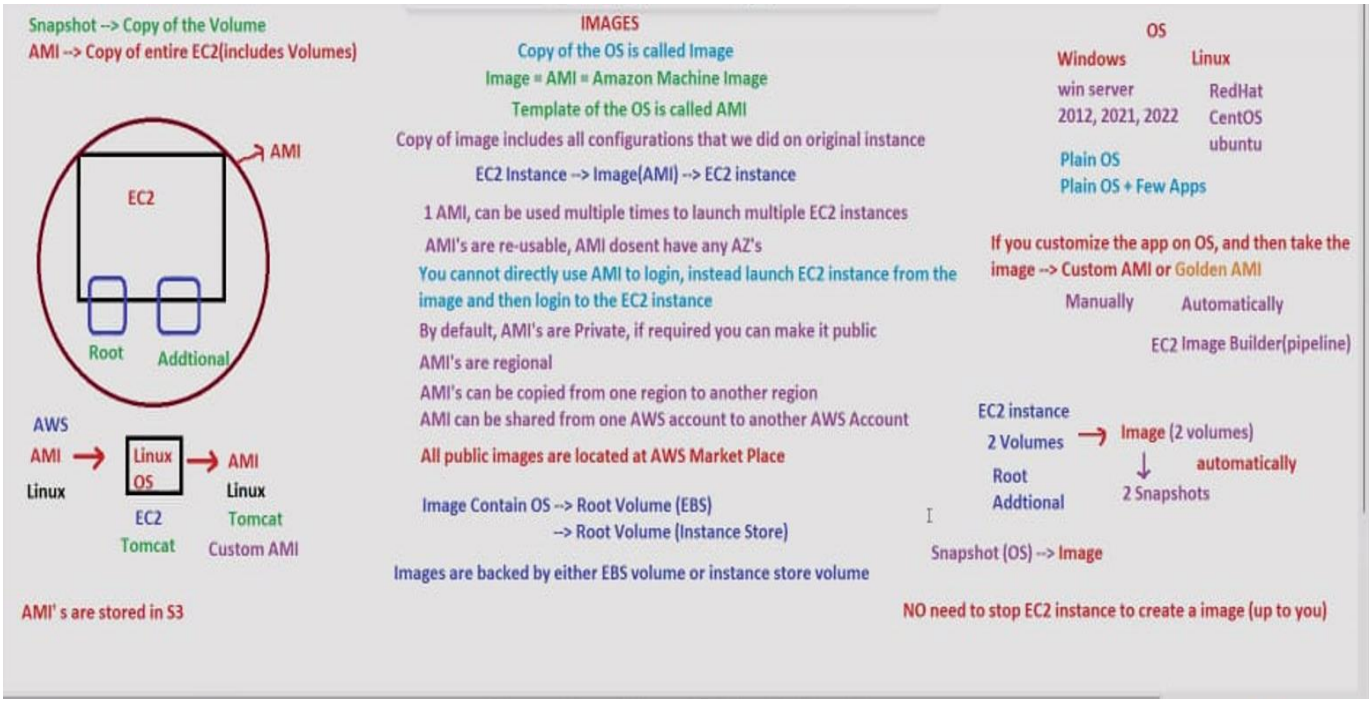
NOT Encrypted --> Encrypt (Copy Option)

All Ecryption Keys are stored in KMS (Key Management Service)

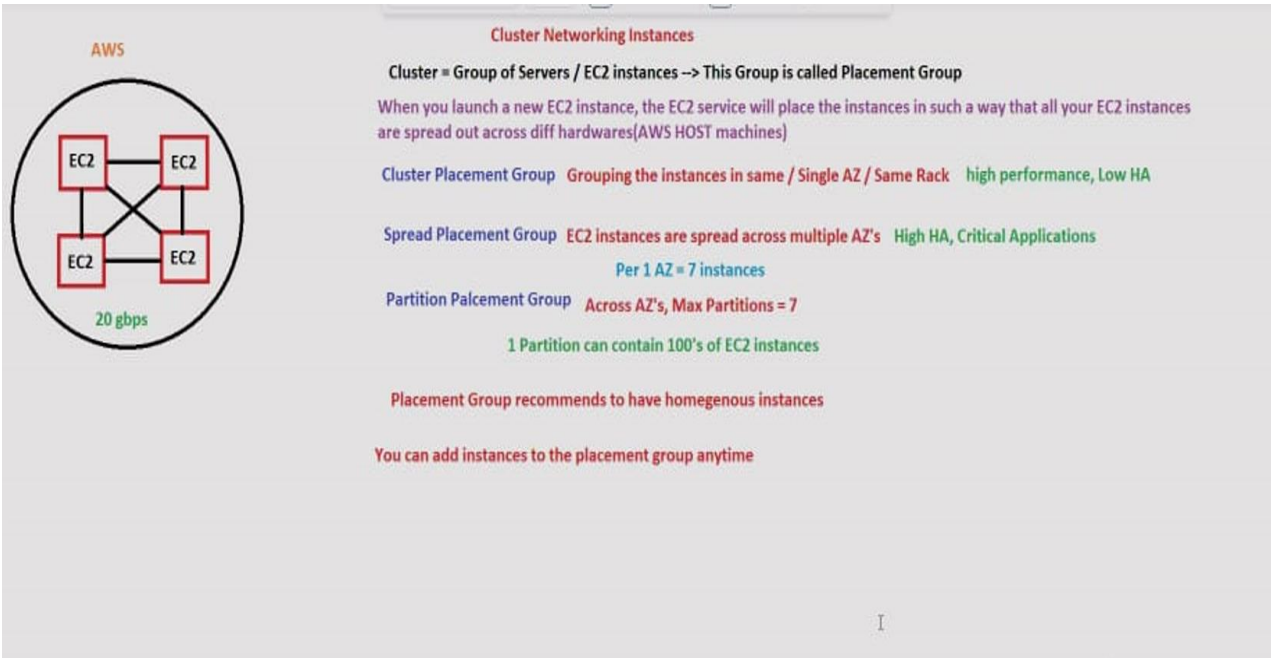
Encryption and Decryption are completely managed by AWS


How to move a volume
from 1AZ to another AZ
1a
RRR Instance
EC2
1a
RRR
1a
CROSS TASK
1b
KGF Instance
EC2
1b
KGF
1b
Snapshot(RRR)
Volume(RRR)
1b
Snapshot(KGF)
Volume (KGF)
1a

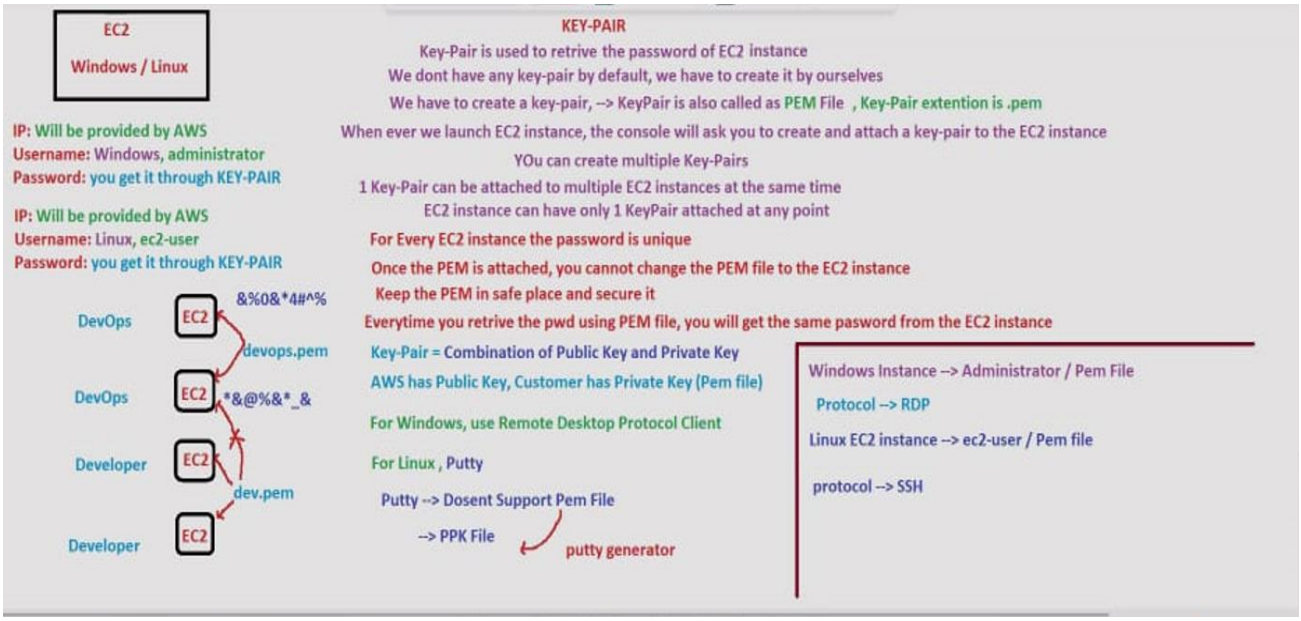
IMAGES

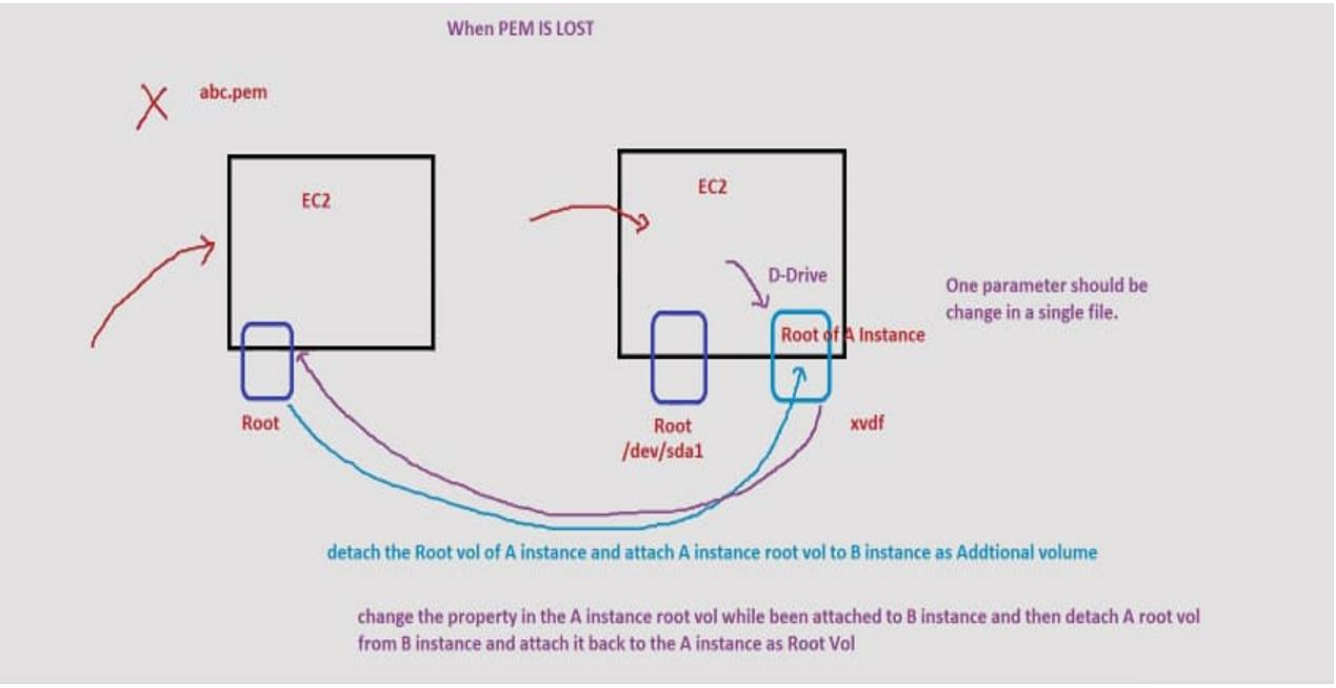

Snapshot --> Copy of the Volume
AMI --> Copy of entire EC2(includes Volumes)
AMI
EC2
Root
Addtional
AWS
AMI
Linux
Linux
OS
EC2
Tomcat
AMI
Linux
Tomcat
Custom AMI
AMI' s are stored in S3
IMAGES
Copy of the OS is called Image
Image = AMI = Amazon Machine Image
Template of the OS is called AMI
Copy of image includes all configurations that we did on original instance
EC2 Instance --> Image(AMI) --> EC2 instance
1 AMI, can be used multiple times to launch multiple EC2 instances
AMI's are re-usable, AMI dosent have any AZ's
You cannot directly use AMI to login, instead launch EC2 instance from the image and then login to the EC2 instance
By default, AMI's are Private, if required you can make it public
AMI's are regional
AMI's can be copied from one region to another region
AMI can be shared from one AWS account to another AWS Account
All public images are located at AWS Market Place
Image Contain OS --> Root Volume (EBS)
--> Root Volume (Instance Store)
Images are backed by either EBS volume or instance store volume
OS
Windows
win server
2012, 2021, 2022
Linux
RedHat
CentOS
ubuntu
Plain OS
Plain OS + Few Apps
If you customize the app on OS, and then take the image --> Custom AMI or Golden AMI
Manually
Automatically
EC2 Image Builder(pipeline)
EC2 instance
2 Volumes
Root
Addtional
Image (2 volumes)
automatically
2 Snapshots
Snapshot (OS) --> Image
NO need to stop EC2 instance to create a image (up to you)

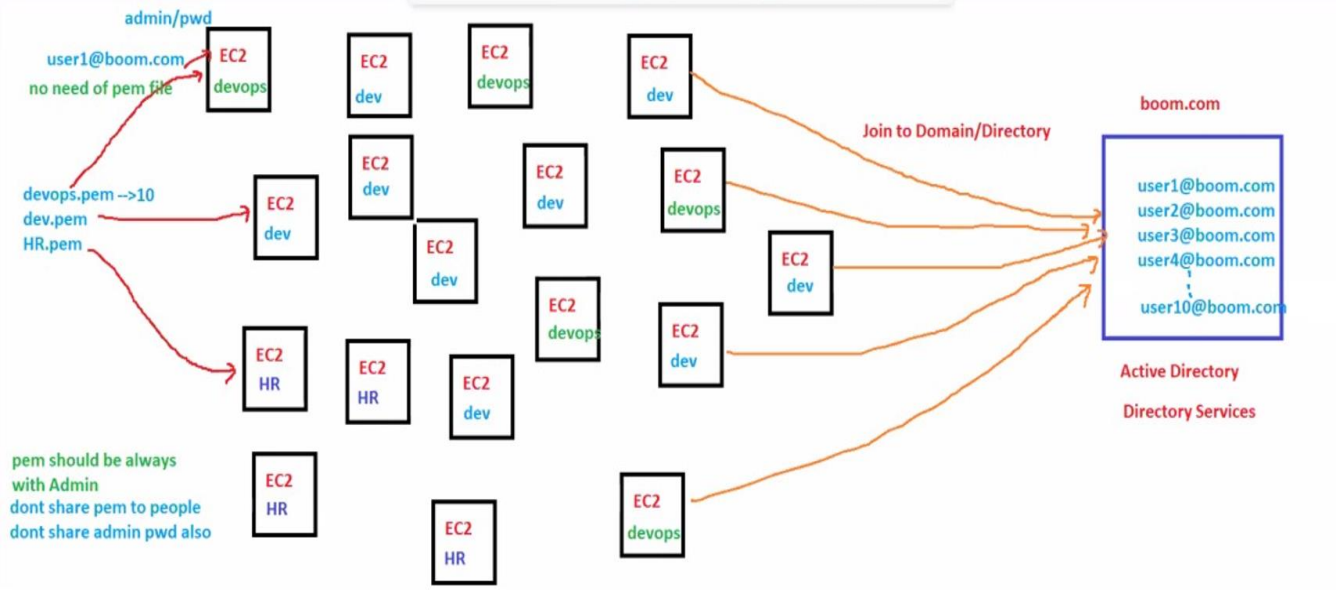
Cluster Networking Instances



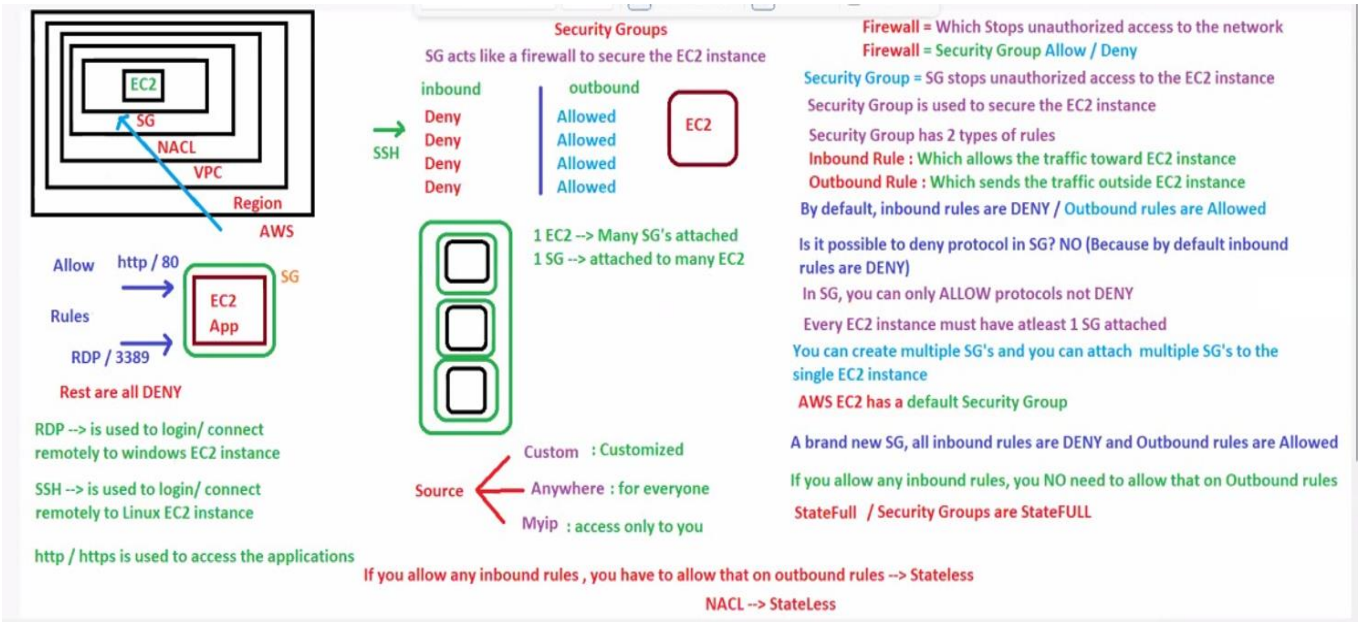
KEY-PAIR

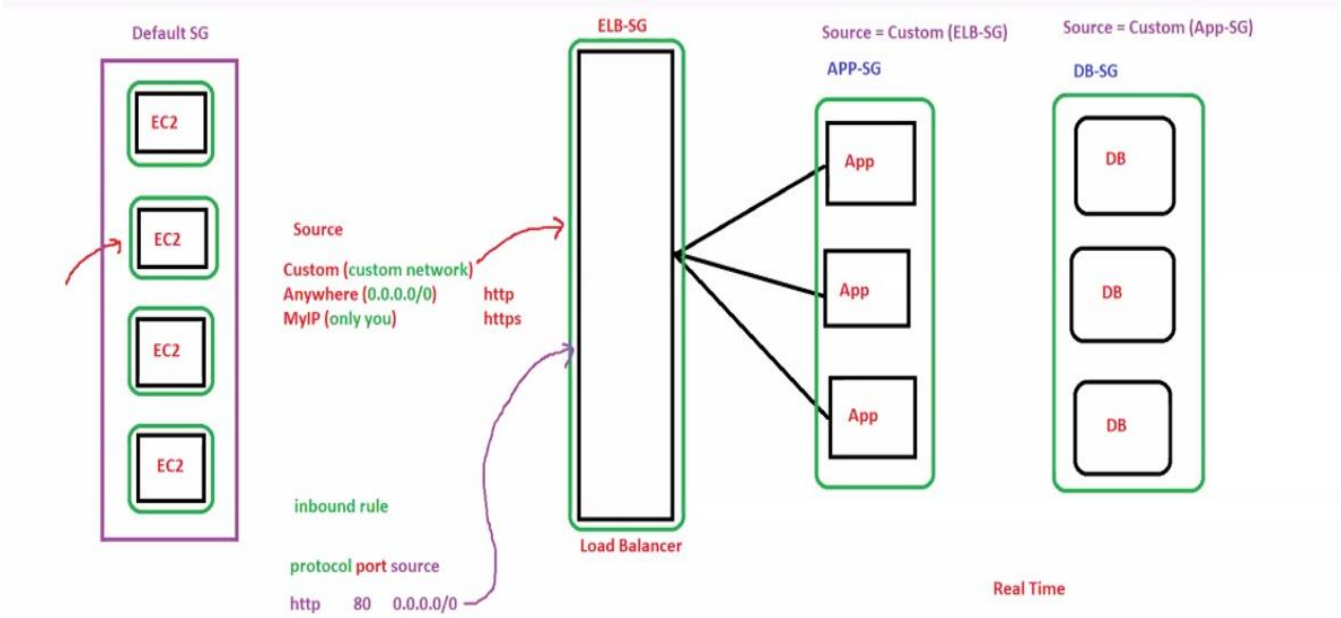



When PEM IS LOST
abc.pem
EC2
EC2
D-Drive
Root of A Instance
One parameter should be change in a single file.
Root
Root
/dev/sda1
xvdf
detach the Root vol of A instance and attach A instance root vol to B instance as Addtional volume
change the property in the A instance root vol while been attached to B instance and then detach A root vol from B instance and attach it back to the A instance as Root Vol

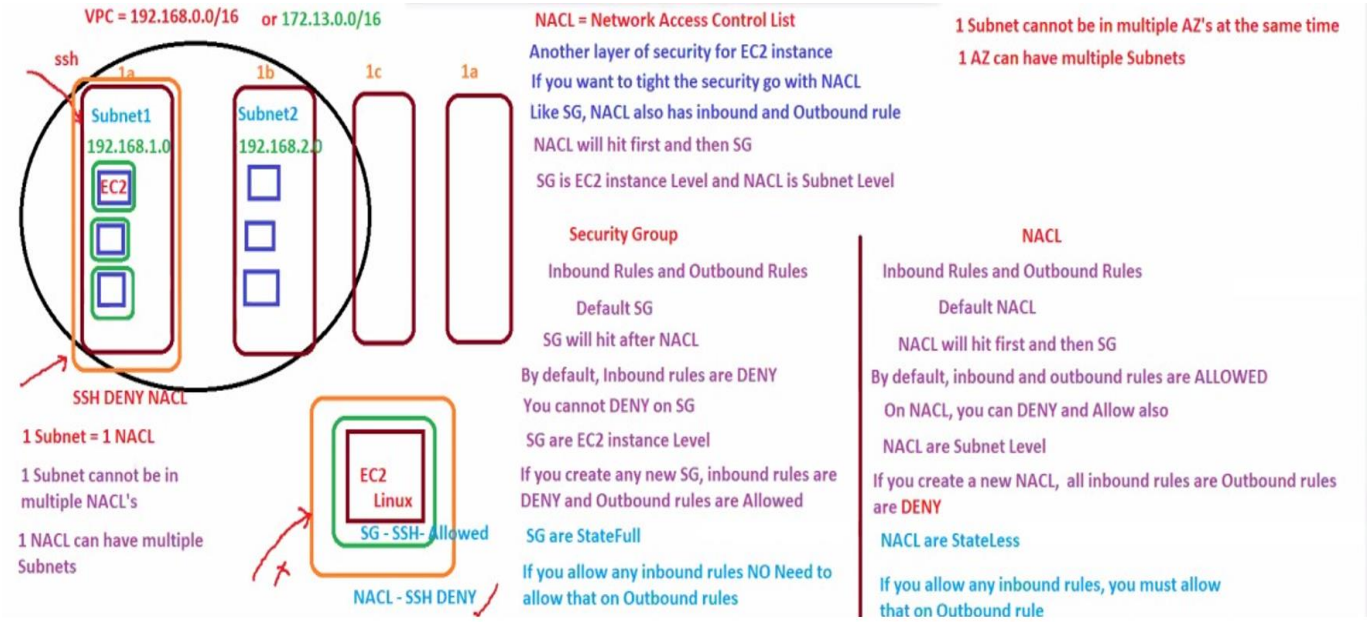

admin/pwd
user1@boom.com
no need of pem file
devops.pem -->10
dev.pem
HR.pem
EC2 devops
EC2 dev
EC2 HR
pem should be always with Admin
dont share pem to people
dont share admin pwd also
Join to Domain/Directory
boom.com
user1@boom.com
user2@boom.com
user3@boom.com
user4@boom.com
user10@boom.com
Active Directory
Directory Services

Security Groups

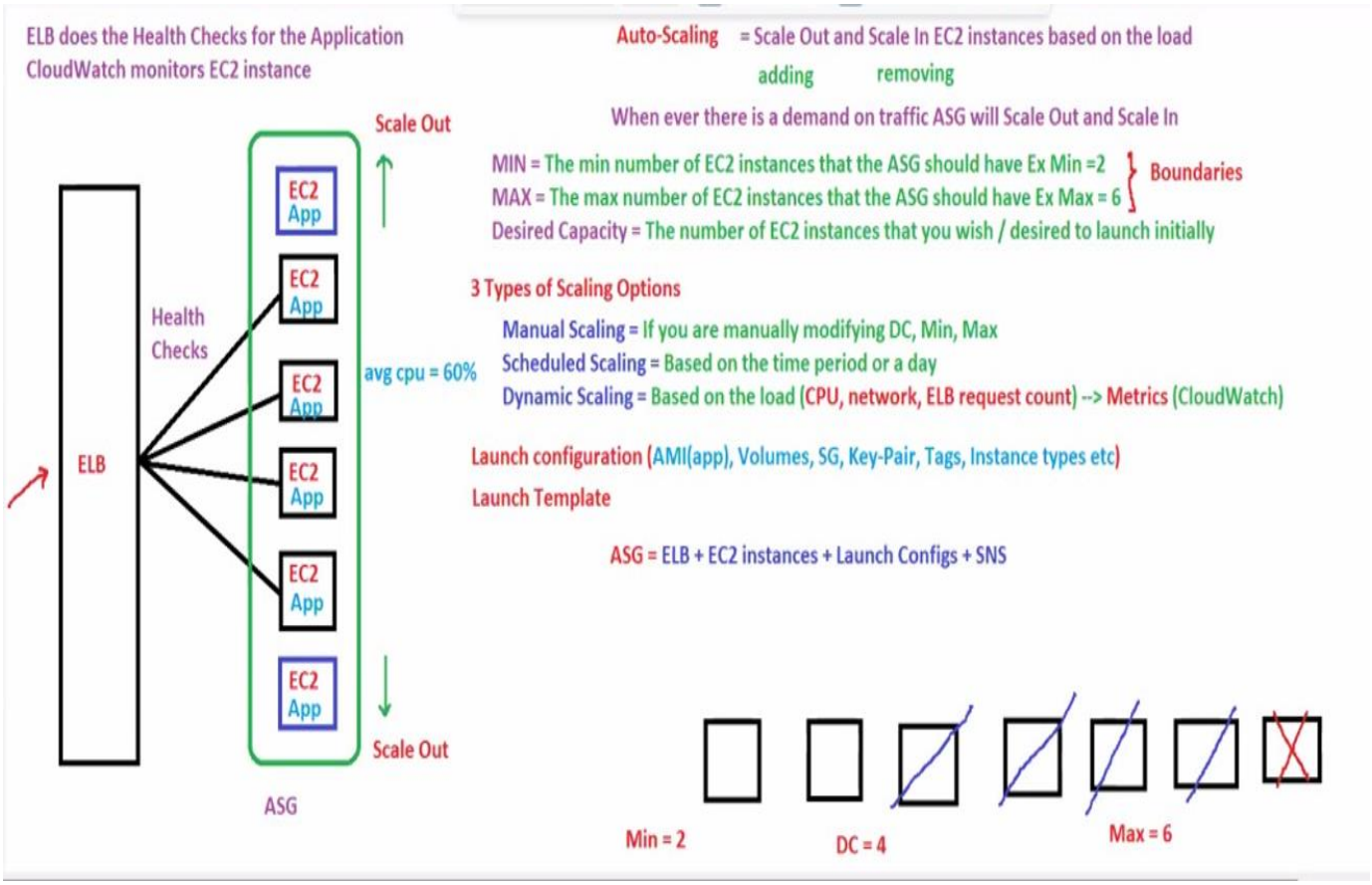




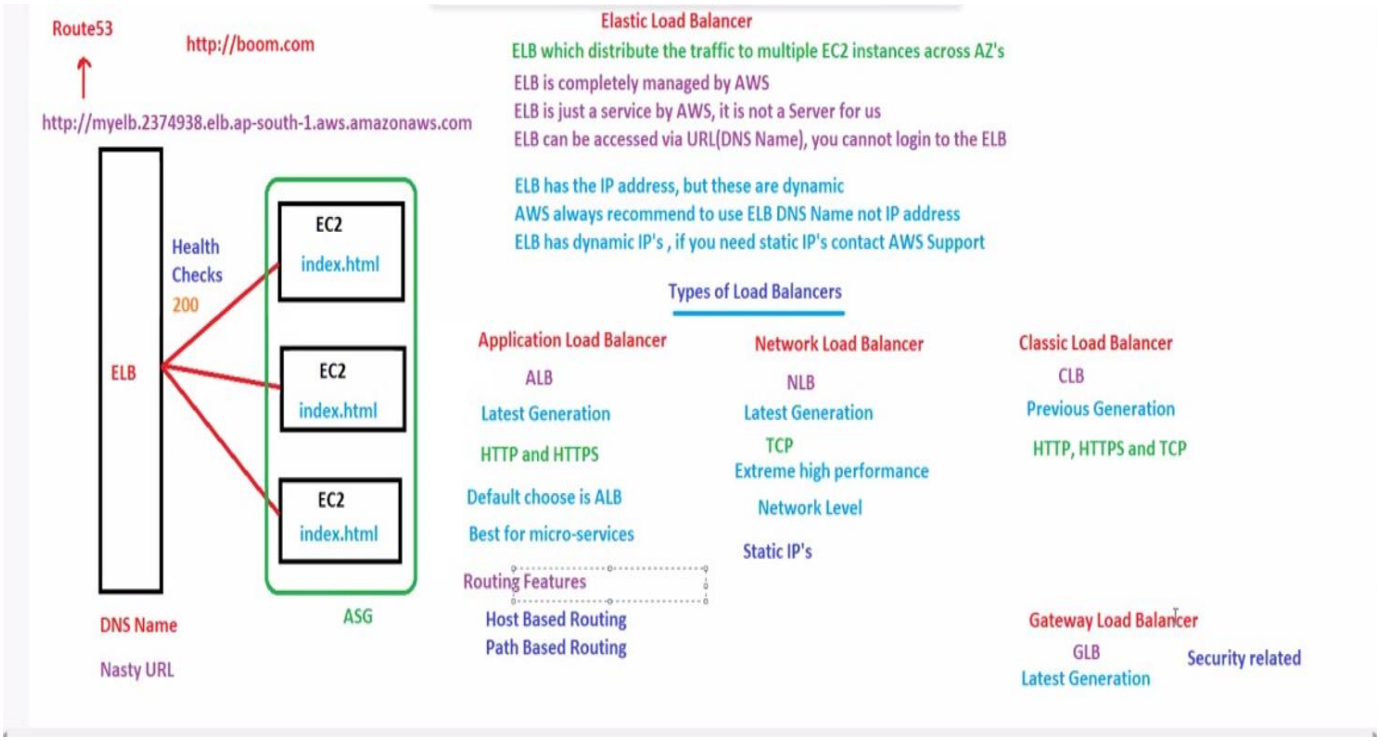
NACL

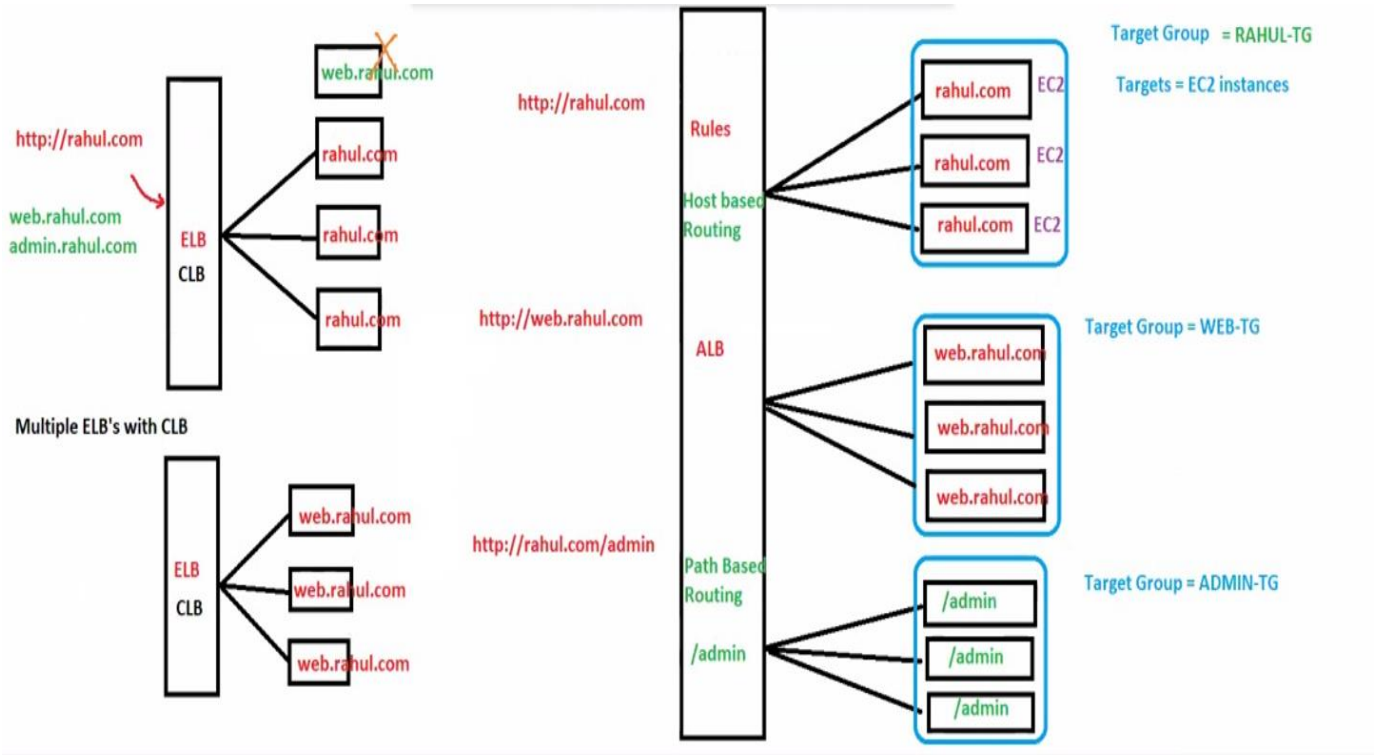


Auto-Scaling :



Elastic Load Balancer :





Types of IP's

Public IP

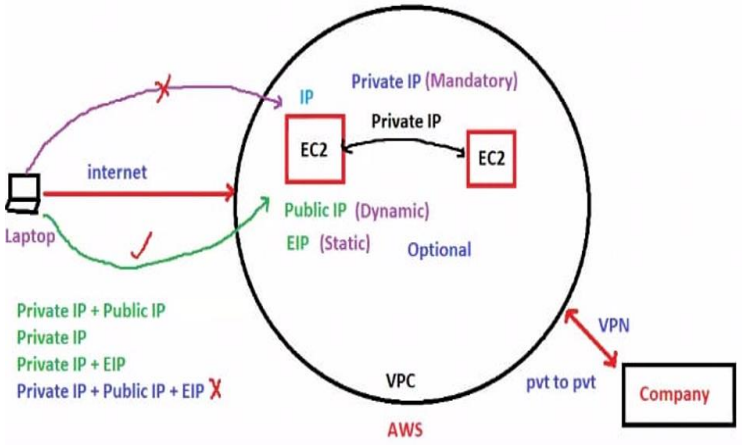
Which can be accessed from internet
Public IP is Optional
Public IP is DYnamic
If you STOP and START EC2 instance , Public IP will be changed.
AWS assigns Public IP's to the instances

Private IP

Which cannot be accessed from the internet
Private IP is mandatory
This is used With in the VPC

Elastic IP

Same as Public IP, EIP is Static
If you STOP and START the EC2 instance, EIP will not change
5 EIP"s are FREE
EIP has to be associated to the EC2 instance
If you have not associated EIP to any instance, it will be charged
If those EIP's are even below 5 also, Dont keep it idle
Idle EIP's are Charged



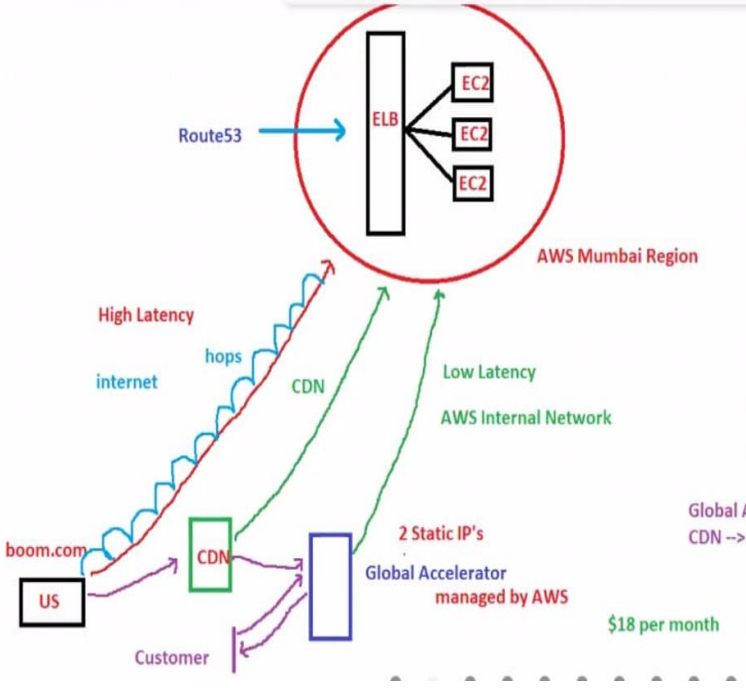
Instance Meta-Data : Data about Instance
From Console, you can get the instance metadata from Description Section
http://169.254.169.254/latest/meta-data/ ---> CLI

User-Data : Boot Strap Scripting
The Script which you provide will run at the boot time of the EC2 instance
Linux --> Shell Script
Windows --> PowerShell

1. AMI 2. Instance Type 3. Configuring 4. Storage 5. Security Group 6. Tags 7. Review / KEY-PAIR

User-data
Shell Script
PowerShell

EC2
tomcat
httpd
any patch
any package



$18 per month

Customer --> CDN --> GA --> ELB
Customer --> GA --> ELB

Global Accelerator --> Will not Cache , but provide static ip's
CDN --> Cache , dosent provide static ip's

Practicle

6 State

1) Pending-Running

2) Stopping-stopped

3) Shuting down to terminating

17-5-2022

TASKS:

1. Launch Windows 2019 EC2 Instance(step 1 to step7(create a key pair and save it safe))

Add a new rule RDP protocal to the Security Group (dont do any changes to the default rule)

use remote desktop protocal to connect to windows machine(username: administrator, pwd: key-pair)

2. Launch Linux redhat EC2 Instance and try to connect it

Download Putty

Download PuttyGenerator

convert PEM to PPK using PuttyGen

--> Open PuttyGen

--> Load your Pem file (Load --> Choose Pem file --> save private key as xxxx.ppk)

Add a new rule SSH protocal to the Security Group(dont do any changes to the default rule)

Use Putty to connect to Linux machine(hostname: publicIP --> expand SSH --> select Auth --> browse ppk file) --> Connect it

3. Try to stop and start and reboot EC2 instance --> public IP changes or not

4. Assign EIP to the machine and try to stop and start --> EIP will change or not

5. Disssociate EIP from the Instance and release it to AWS.

6. Create Launch template and launch EC2 instances from it and then and delete it (launch ec2 isntances from template)

7. Try shutdown behaviour from ec2 instance and see if the instance is stopped or terminated, change to terminate and then shutdown again and see the difference.

8. Review options in ACTIONS

9. Terminate EC2 instances.(if you have protection enabled --> disable that first from instance settings --> change termiantion protection)

18-5-2-22

****

IMAGES TASKS

****

Launch a windows instance -->login to that instance --> create some files or install a software

--> Create image --> launch instance from image --> login to the newly created instance(password?) --> Check the files or software are availble

Copy the image to another ireland region --> once verified, delete the image in ireland region

Encrypt the image --> use copy option --> check encryption option --> create image --> launch ec2 instance --> login and see the difference

Copy the encrypted image to same region and another region (?)

Copy the Snapshot to another region --> once verified ,delete the snapshot

Encrypt the Snapshot and share it to the other AWS account ?

Try share the image and snapshot to another account

Create a image from the snapshot

First try deleting the snapshot and then AMI --> see if this works

First try deleting the AMI and then snapshot--> see if this works

Archive snapshot and restore it again

19-05-2022

TASKS

****

1. CROSS TASK (additional volumes + Root Volumes) - different AZ's

2. Increase the root volume size

3. Add addtional volume and make it available to the user.

4. Play with attach and detach volume for Root and Addtional volumes

   --> detach 1a root volume and attach it to other new 1a EC2 instance as root vol (same AZ)

   --> detach 1a root vol and attach it to other new 1a EC2 instance as addtional vol (same AZ)

--> Assume you lost your pem file, and we need to login to the EC2 instance how?

--> Take a snapshot (Mumbai region) and copy to another region(ireland region) --> from snapshot --> create volume --> attach it to ec2 instance(ireland)

5. Extend the existing volume on Linux

6. New addtional volume for Linux

7. Life Cycle Manager policy and delete it --> make sure snapshots are created using UTC time (tag volumes properly)

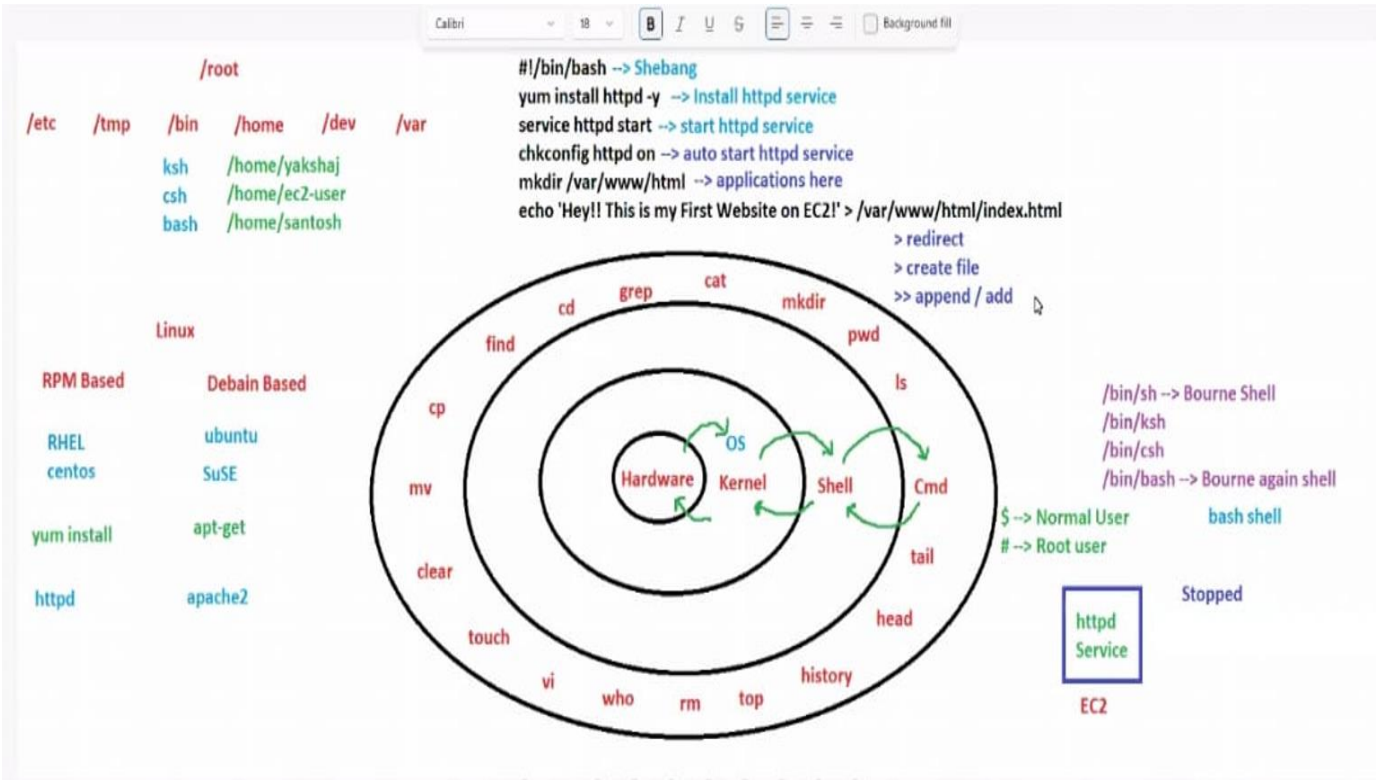
8. Terminate EC2 instances

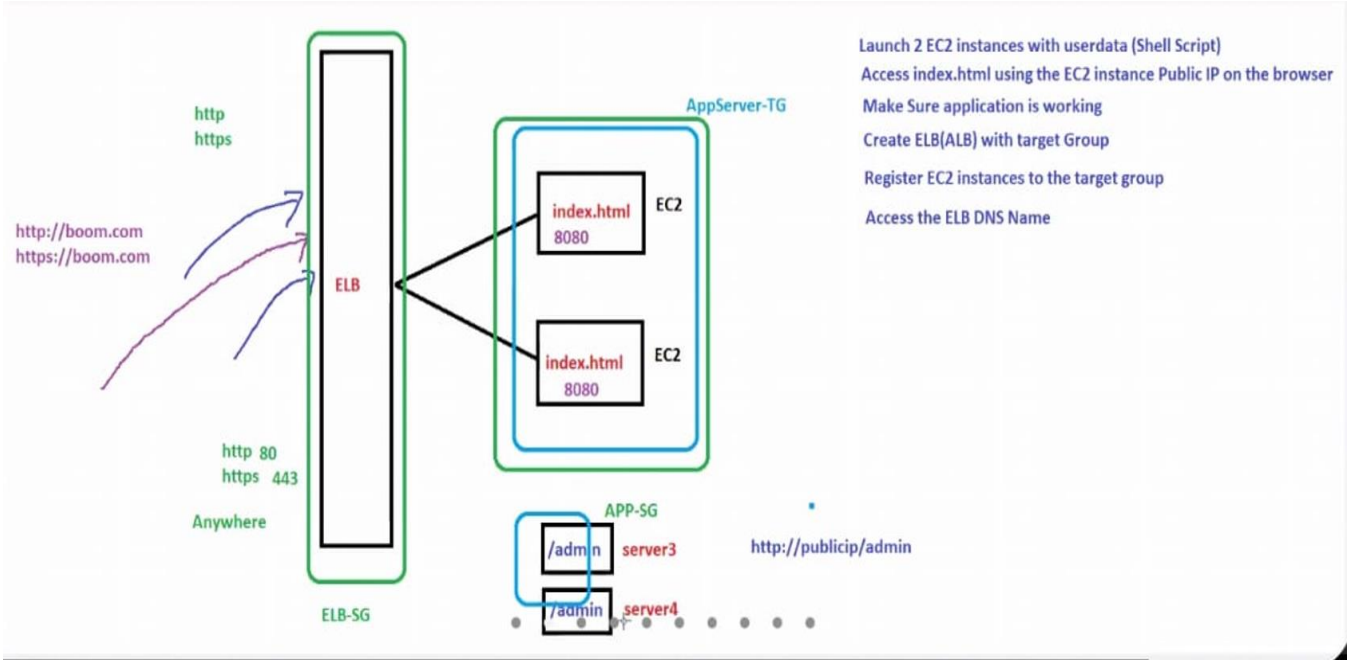
9. Delete additional volumes, life cycle policy

Extra URL:

https://youtu.be/FUXy2t28oqc

20-05-2022





#!/bin/bash

yum install httpd -y

service httpd start

chkconfig httpd on

mkdir /var/www/html

echo 'Hey!! This is my First Website on EC2!' > /var/www/html/index.html

**>>>

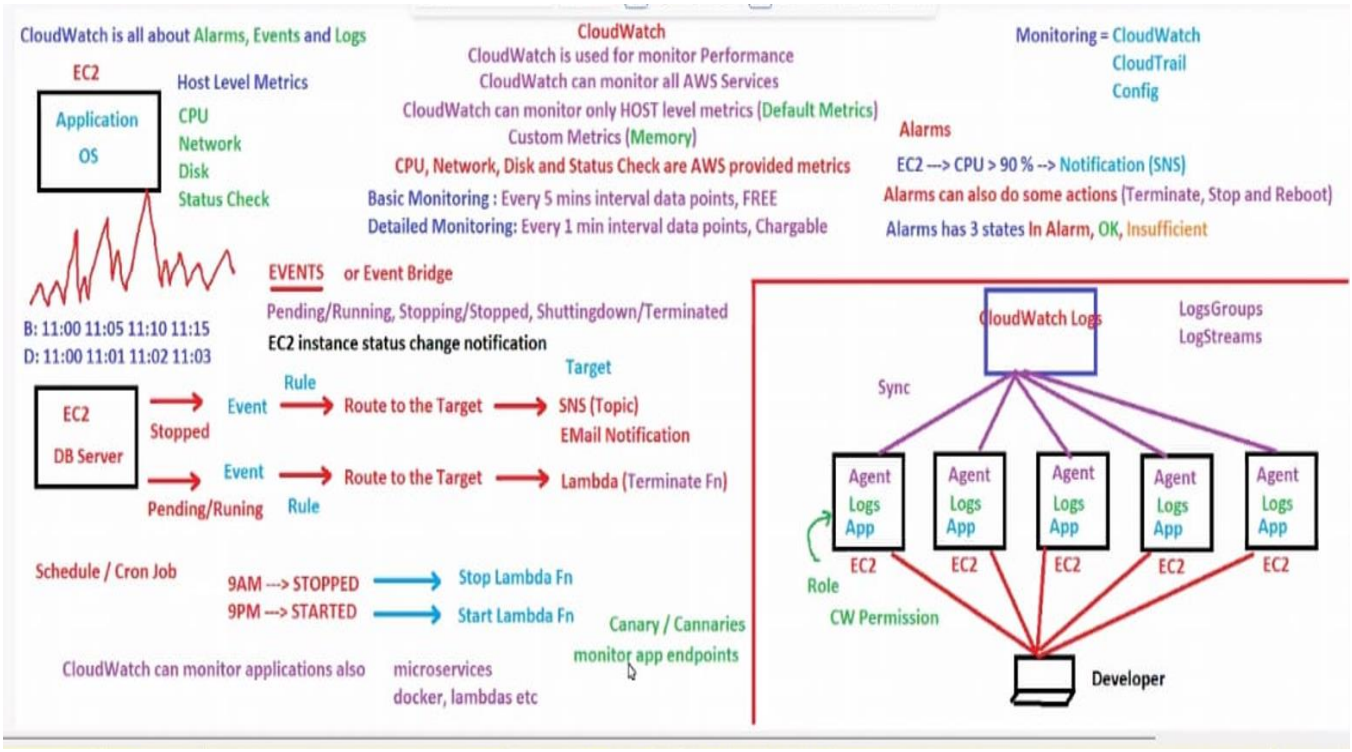
```
6. Check the targets in target groups, it should show healthy --> try stickiness
   --> configure stickyness
7. Explore path based routing (create 4 instances, 2 with /index.html(TG1) and 2 with /admin(TG2) and add rules in tar
7. Terminate EC2 instance, ELB and target groups.


**************************
Auto-Scaling
**************************
1. Create ELB with Empty Target Group
1.1: Keep HTTP in your SG, and make sure you have default rule ALL TRAFFIC with self
2. Create Launch Configurations(select AMI ID, provide user data script)
3. Create ASG from the created Launch configuration (min, DC, max)
3.1: Check your targets in target groups, it should be healthy
4. Access your ELB DNS Name.
--> try to do manual scaling
--> try to do schedule scaling
--> try to do instance refresh
--> try to terminate ec2 instances manually and see what happens
5. Delete ASG
6. Delete launch Config
7. Delete ELB and TG
Explore: Setup an application inside EC2 --> Create AMI --> Use this Custom AMI in LC and create ASG using this LC.
update the application again --> Create new AMI --> Copy and Create LC with new AMI --> Update ASG with new LC
```

While Creating the image you can add the number of additional volumnes

Encrypted snapshot can not be shared to others

23-05-2022



24-5-2022

(Practise on Cloudwatch)

==========

Create a role with cloudwatchfull access

launch Amazonlinux2 ec2 instance and attach role

sudo yum update -y --> it will update all patches

sudo yum install -y awslogs --> installing CW agent

Edit the file and check the log path: /etc/awslogs/awslogs.conf

Edit region /etc/awslogs/awscli.conf

sudo systemctl start awslogsd

Check Cloudwatch Logs

****

CloudWatch

****

--> Stop or Terminate the EC2 instance by creating Alarm (if the CPU is less than 10%)

--> Create Alarm from CloudWatch Console

--> Create a billing alarm (> 100 INR)

--> Create an Event and get the Email notification if any one launch the EC2 instance

--> Try Schedular in CloudWatch with Lambda (Stop and Start EC2 instance) --> Follow URL

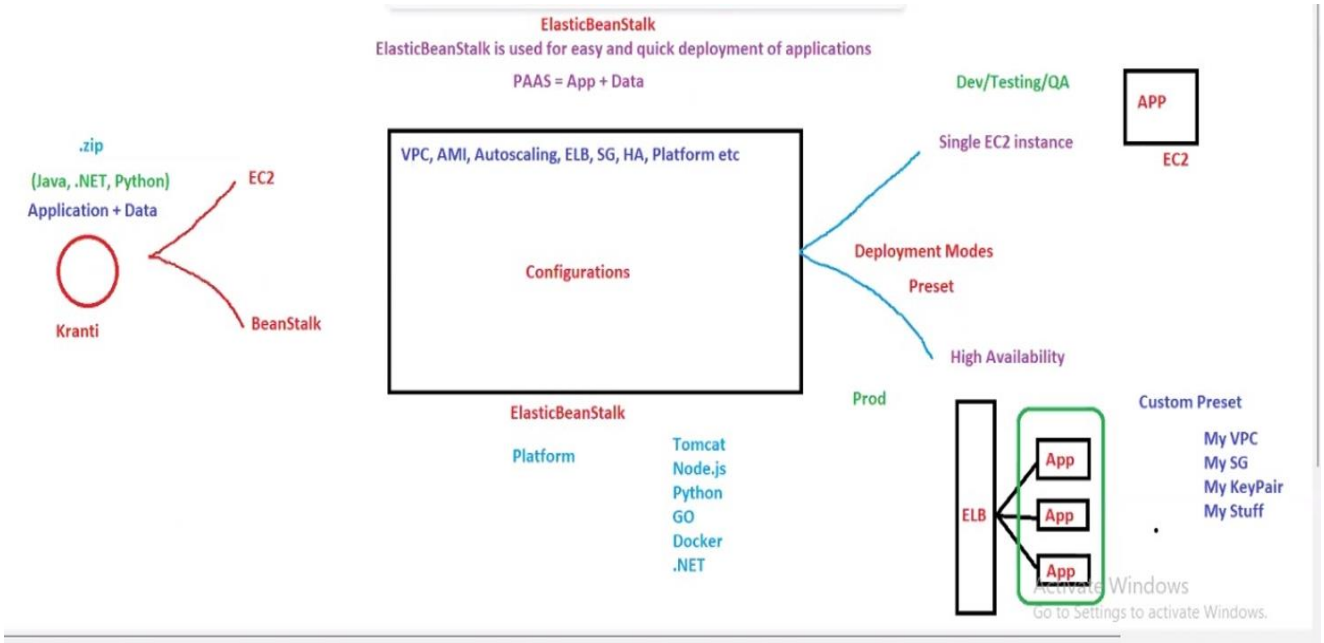
--> Setup CloudWatch Logs on Linux

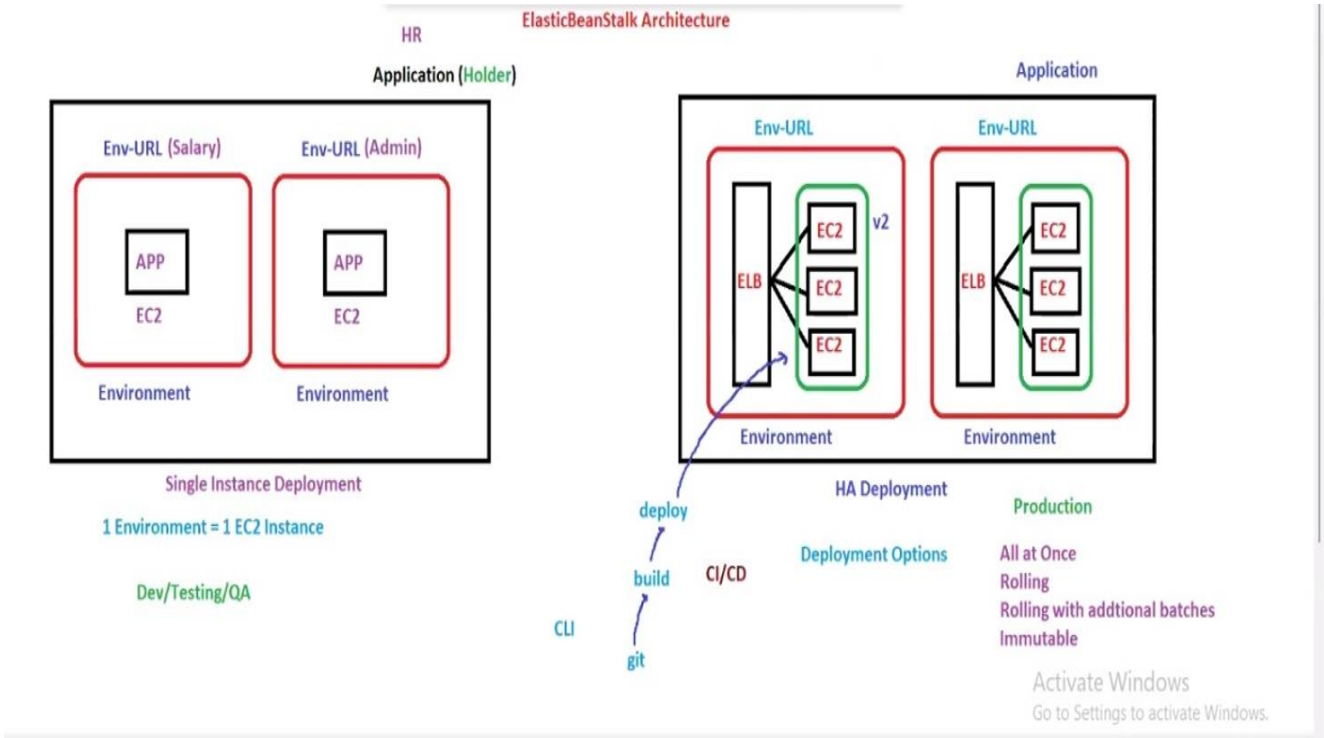
--> Create a sample Dashboard and add widgets

--> Clean up Everything(Terminate EC2, delete dashboard,events,Logs, Cannary,S3 bucket)

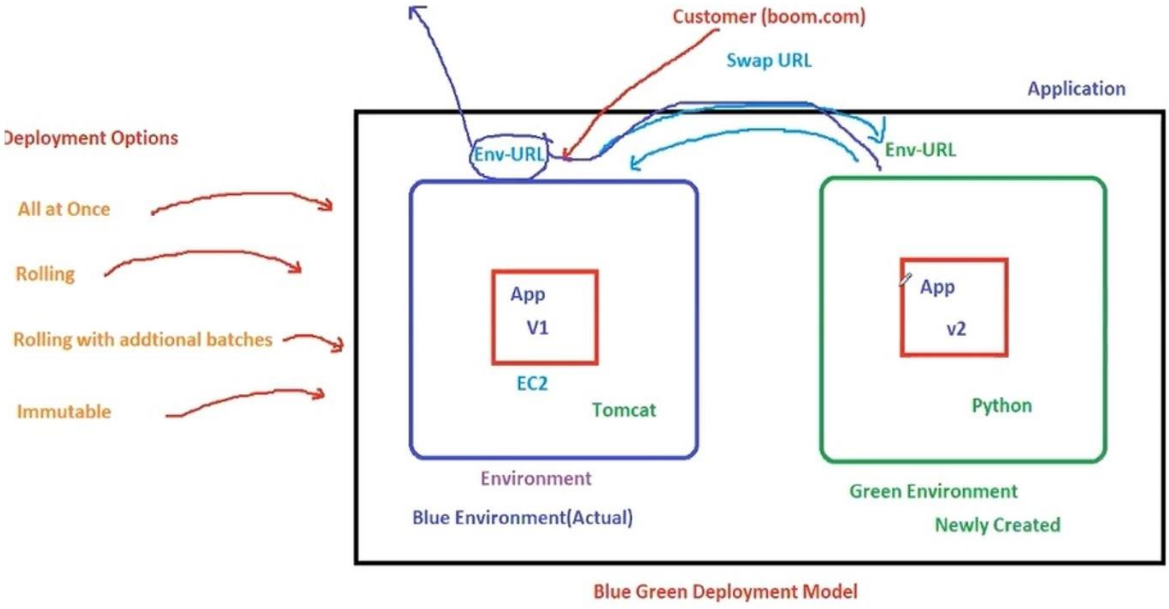
--> Setup CloudWatch Logs on Windows(explore) + Custom memory metrics

25-5-2022


ElasticBeanStalk
ElasticBeanStalk is used for easy and quick deployment of applications
PAAS = App + Data
.zip
(Java, .NET, Python)
Application + Data
Kranti
EC2
BeanStalk
VPC, AMI, Autoscaling, ELB, SG, HA, Platform etc
Configurations
ElasticBeanStalk
Platform
Tomcat
Node.js
Python
GO
Docker
.NET
Dev/Testing/QA
Single EC2 instance
APP
EC2
Deployment Modes
Preset
High Availability
Prod
ELB
App
App
App
Custom Preset
My VPC
My SG
My KeyPair
My Stuff


ElasticBeanStalk Architecture
HR
Application (Holder)
Env-URL (Salary)
Env-URL (Admin)
APP
EC2
APP
EC2
Environment
Environment
Single Instance Deployment
1 Environment = 1 EC2 Instance
Dev/Testing/QA
Application
Env-URL
Env-URL
ELB
EC2
EC2
EC2
v2
ELB
EC2
EC2
EC2
Environment
Environment
HA Deployment
Production
deploy
build
CI/CD
CLI
git
Deployment Options
All at Once
Rolling
Rolling with addtional batches
Immutable

26-05-2022



Blue Green Deployment Model

****
BeanStalk
****
--> Create Application with single instance preset
--> Create beanstalk env with tomcat --> Access the website using Env-URL
--> Create beanstalk env with Python--> Access the website using Env-URL
--> GO to application level options Swap the URL and then access the env-URL (Blue Green Deployment)

--> Create a new Application and environment with HA preset and check in EC2 console for ASG, ELB, TG etc
--> Clone env, restart env, rebuild env ---> See what is happening
--> do Blue green deployment on HA
--> Terminate the environment and restore it
--> Download the logs and Go through it once
--> Delete the Application
Upload you own application and redeploy with latest application
****

Activate Windows
Go to Settings to activate Windows.